Chapter 7 Air Quality
CIVL1105 Environmental Engineering



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Expected leaning outcomes for Chapter 7

Major air pollutants (source, effects, control)

Air pollutant phenomena (e.g., photochemical smog, meteorology effect, etc.)

Air pollution models

Atmospheric stability

Great Smog of London (1952)





>4000 deaths

https://www.youtube.com/watch?v=90kq35cWZz4



New York City, United States, in June 2023, smoke from the Canadian wildfires

Air pollution in China

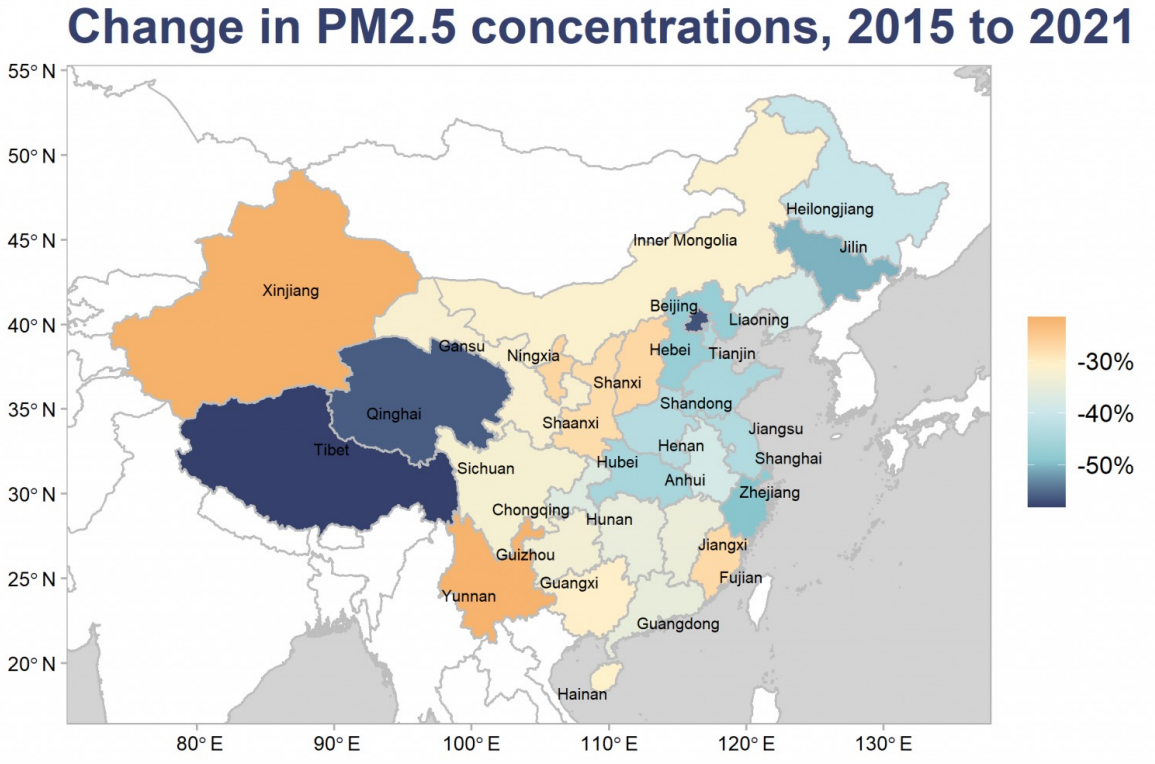


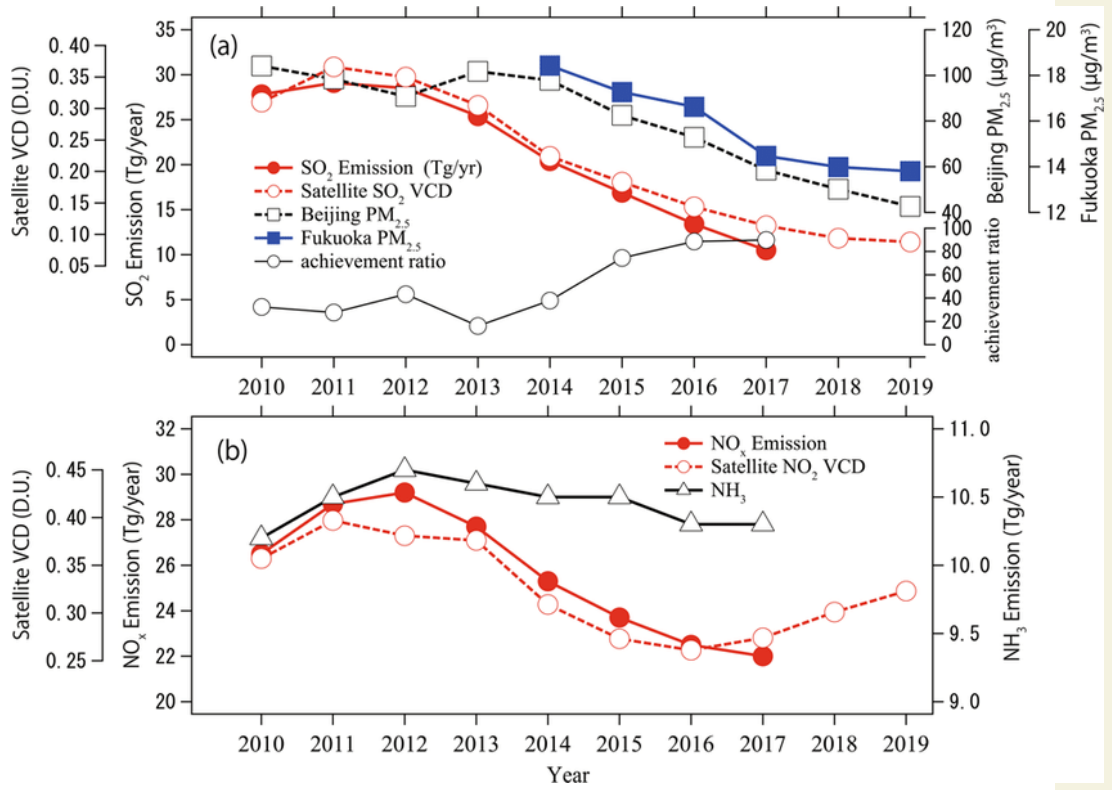


http://www.trust.org/slideshow/?id=36163d90-df85-45e3-ab30-5d97b634bfa4&source=search

Air pollution in China

[Paradigm shift in aerosol chemical composition over regions downwind of China | Scientific Reports (nature.com)]





[Beijing's air quality meets national standards: a major milestone in China's war on smog – Centre for Research on Energy and Clean Air]

Why does Air Pollution matter?

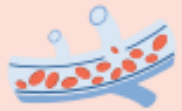
Health effect

- CO poisoning
- PM lung penetration
- Ozone respiratory irritation

Nuisance

- Poor visibility
- Bad smell



Pollutants inhaled into lungs	Transmission to CV system	Impaired CV function	Diseases and exacerbation
Penetration into lungs depends on pollutants and particle size	Neural and endocrine pathways	**Heart** Altered Rhythm increased susceptibility to ischemia	**Cardiovascular morbidity and mortality**
 **Air Pollution**			Accelerated development cardiovascular disease
	Pollutants or their constituents enter the circulation	 **Blood Vessels** Constriction, poor relaxation, stiffening and increased blood pressure	
Inflammation and oxidative stress in lungs	 Transmission of inflammatory and oxidative mediators	 **Blood** Increased coagulation. Decreased fibrinolysis	Triggering of a cardiovascular event (e.g., heart attack or stroke)

Credit: World Heart Report 2024, World Heart Federation.

CV = cardiovascular.

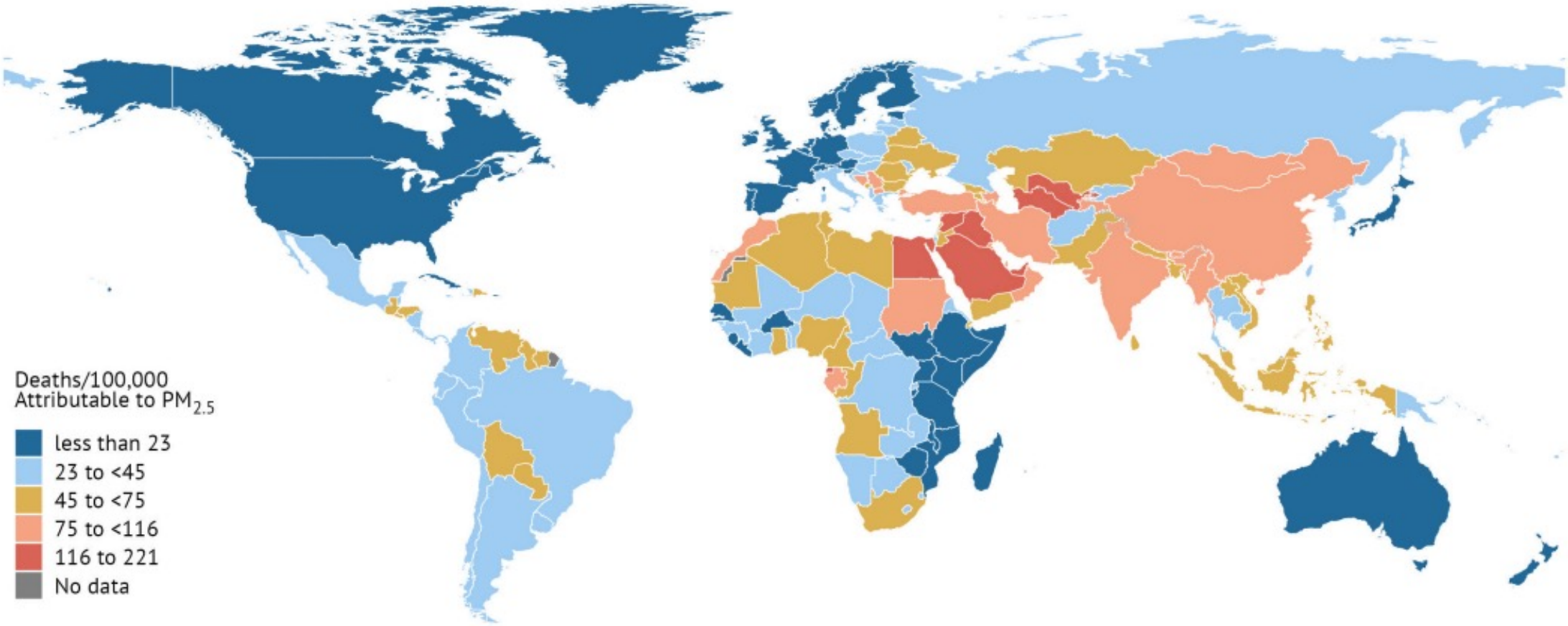


FIGURE 19: Global map of age-standardized rates of death attributable to ambient $PM_{2.5}$ in 2023. The age-standardized death rate is a useful metric for comparing the burden of disease attributable to a particular risk factor across countries or regions because it factors in population age structure and size.

https://www.stateofglobalair.org/resources/report/state-global-air-report-2025

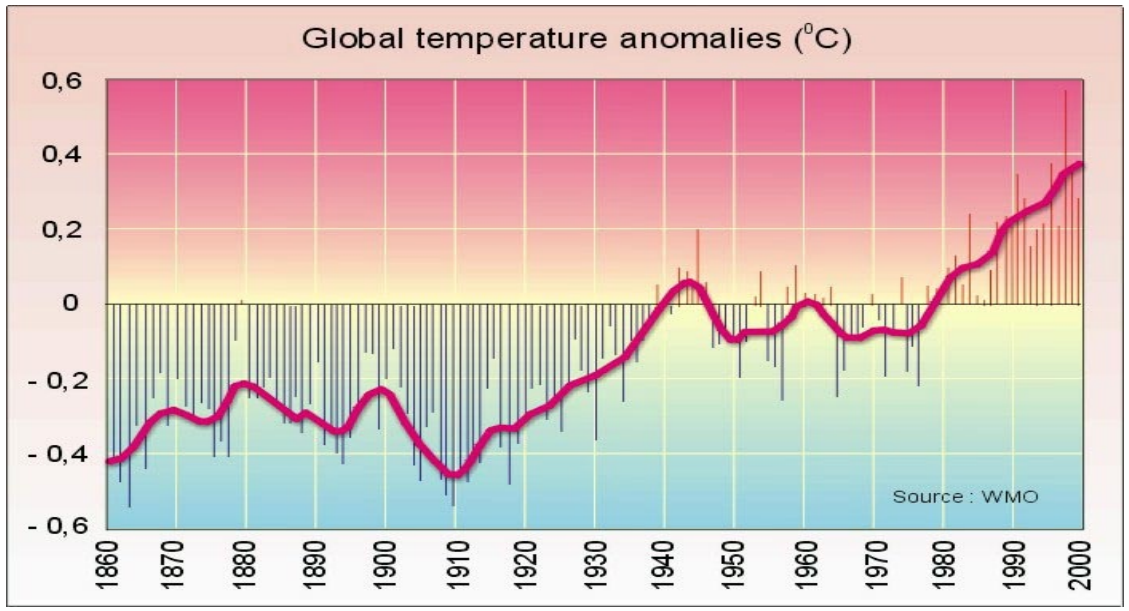
Adverse effects of air pollutants

Damage to vegetation and materials

- Acid rain

Global climate change

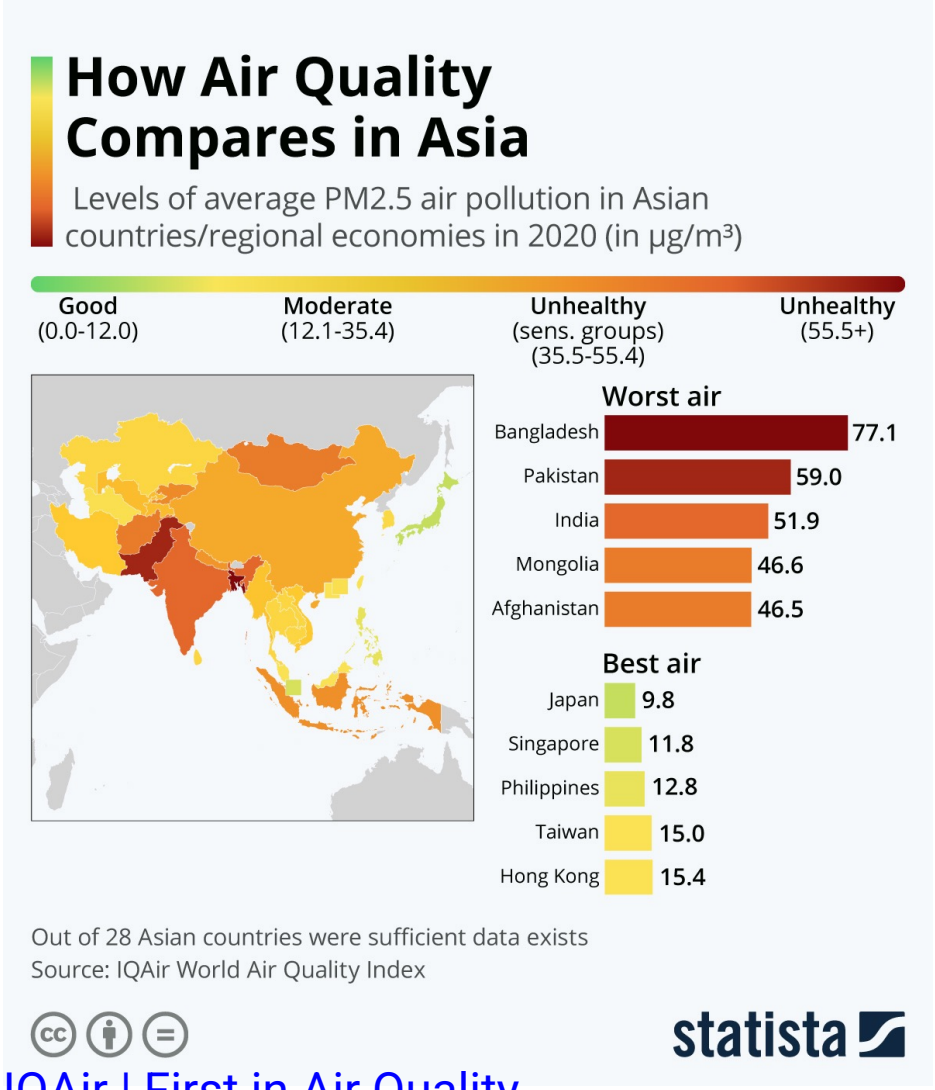
- Global warming (CO_2)

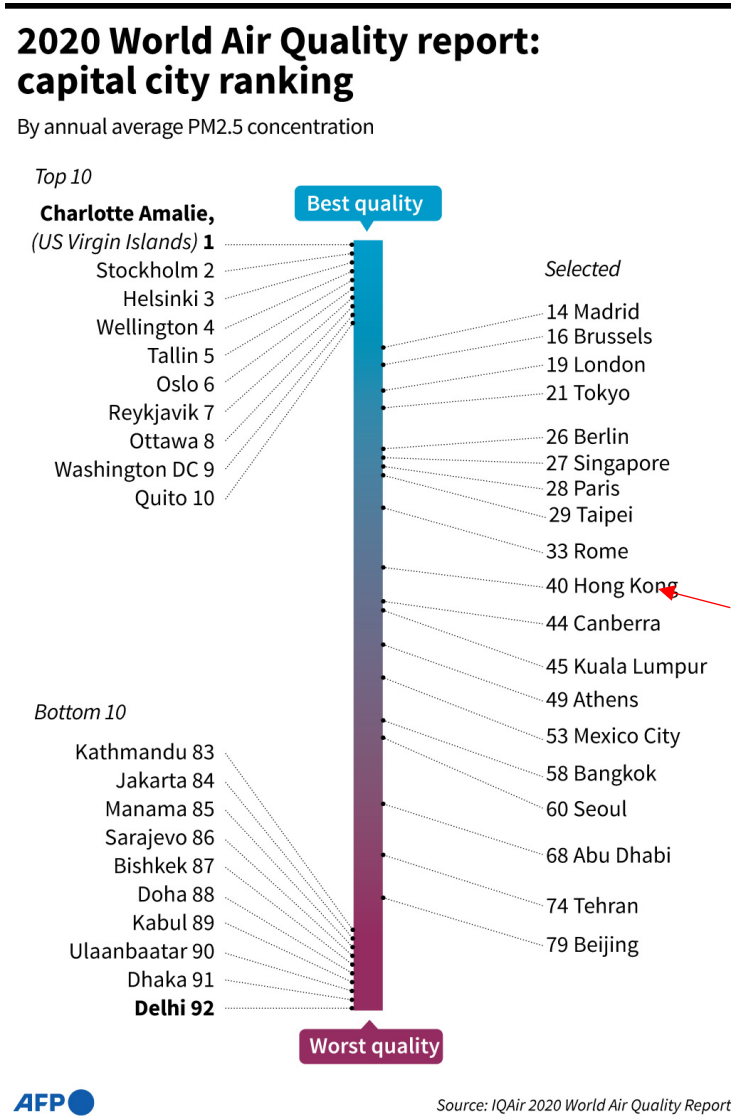




Damage caused by acid rain

Air pollution in Hong Kong

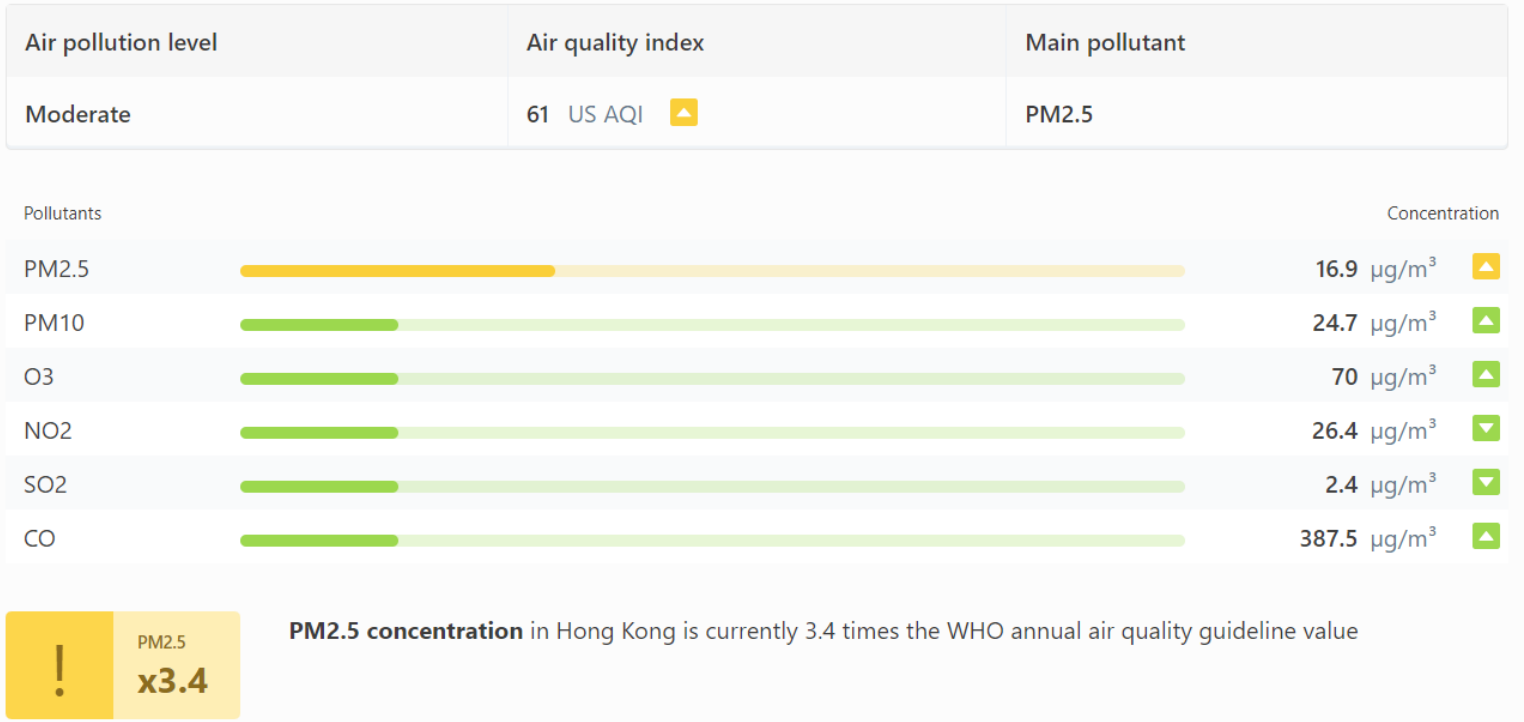




[IQAir | First in Air Quality]

2020 World Air Quality report: capital city ranking

Air pollution in Hong Kong



Air pollution level	Air quality index	Main pollutant
Moderate	61 US AQI	PM2.5

Pollutants	Concentration
PM2.5	16.9 µg/m³
PM10	24.7 µg/m³
O3	70 µg/m³
NO2	26.4 µg/m³
SO2	2.4 µg/m³
CO	387.5 µg/m³

The main sources are motor vehicles, marine vessels and power plants. On the street level, vehicles – especially buses and trucks – are the main culprits for pollution

[Hong Kong SAR Air Quality Index (AQI) and Hong Kong SAR Air Pollution | IQAir]

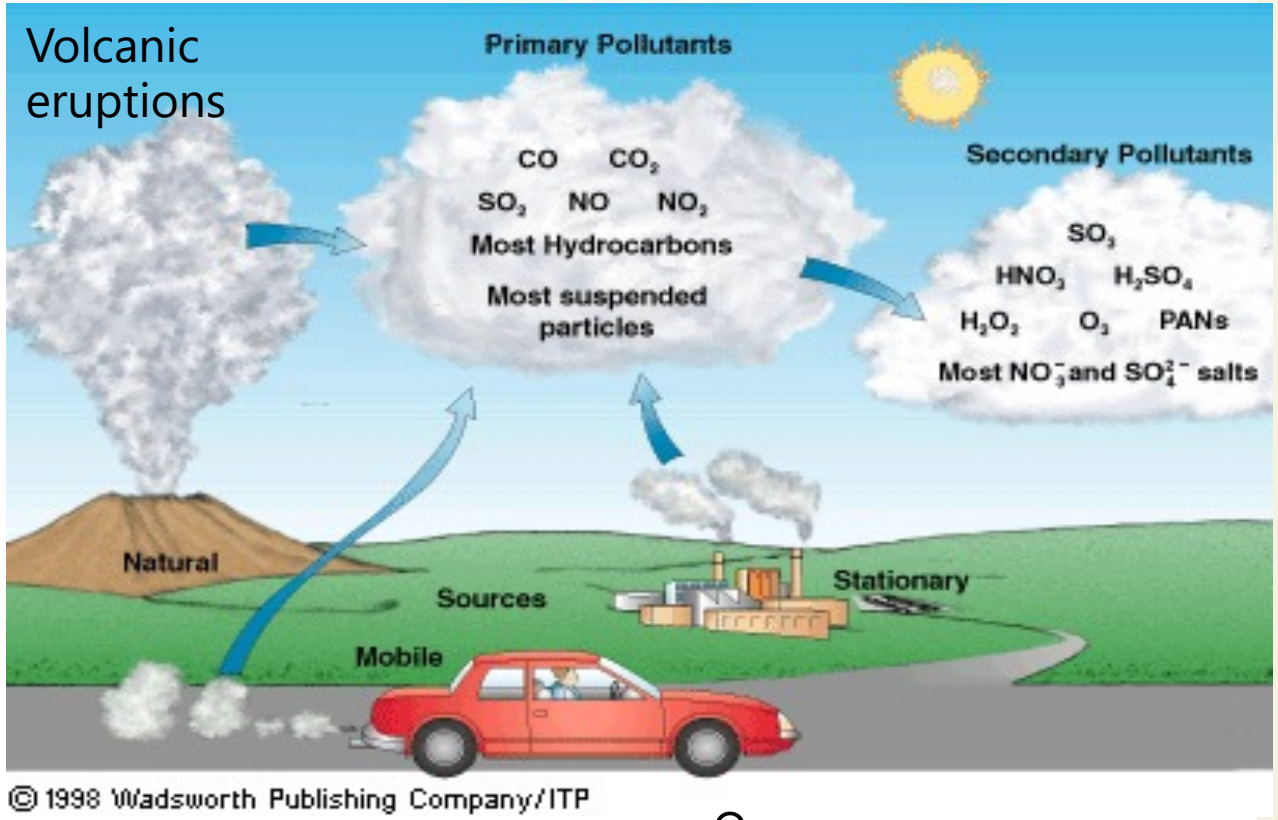
[8 Key Facts About Air Pollution In Hong Kong - Green Queen]

What is air pollution?

Substances that are not part of composition of “air”, which can be harmful to people or their environment.

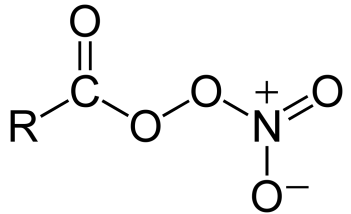
- **Primary Air Pollutants** – directly emitted from the source

- **Secondary Air Pollutants** – formed from reactions between existing air pollutants under certain conditions



Where is the pollutants from?

Peroxyacetyl nitrate (PAN)



Where does it come from?



Combustion of fuel under ideal condition (complete combustion):

Fuel (e.g., CH_4) + O_2 → **CO_2** + 2 H_2O

Incomplete combustion (temperature, oxygen, residence time, mixing,…):

Fuel (e.g., CH_4) + O_2 → mostly (**CO_2** + 2 H_2O) + traces of [**CO + HC**]

Partially combusted hydrocarbons

Nitrogen reacts with oxygen at high temperature:

air(N_2 + O_2) + heat → **Thermal NO_x**

Impurities in fuel (N, S, Pb, Hg, ash, …):

Fuel(H, C, S, N, Pb, Hg, ash) + air(N_2 + O_2) →

CO_2 + H_2O + **CO + HC + NO_x + SO_x + Pb + Hg + particulates + ash**

Where does it come from?

Photochemical reaction:

VOCs + NO_x + sunlight → photochemical smog (O_3 + VOCs etc.)

Organic compounds that easily vaporize

Ground level ozone

Sources:

Mobile source: automobiles, trucks, other transportation → Most CO, nearly half of NOx

Stationary source: power plant, industrial processes, chemical manufacturing, et al. → Most of SOx, Hg, particulates, VOCs, ~> half of NOx

Criteria pollutants

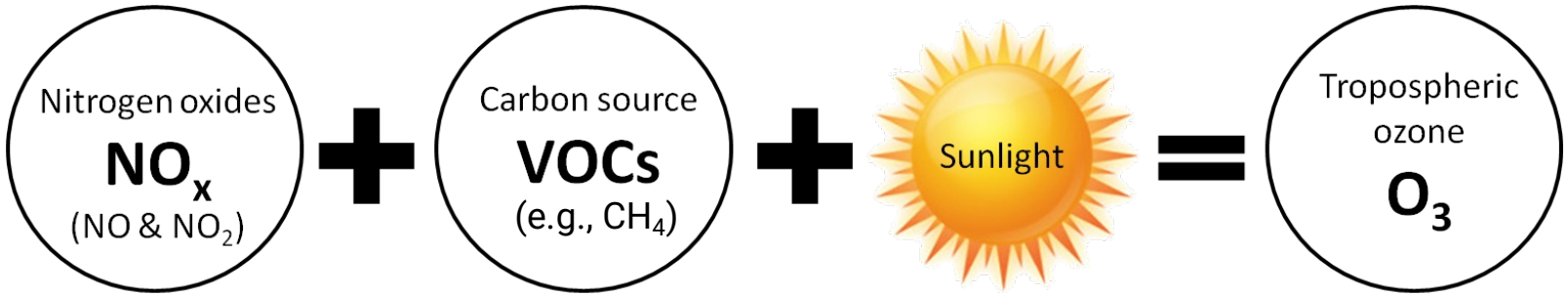
Air pollutants

Primary pollutants

- Pollutants emitted directly into the atmosphere
  - **Combustion (power plant stack gas and automobile exhaust emission)**
  - Evaporation (volatile compounds such gasoline, paints, …)
  - Grinding/abrasion (dust)

Secondary pollutants

- Pollutants created by physical/chemical processes in the atmosphere
- E.g., ground level ozone



https://www.climatesignals.org/climate-signals/surface-ozone-change

Sources of air pollution

Man-made pollutions

- Transportation
- Industrial process (petroleum, cement, etc.)
- Fuel combustion (power plants)
- Solid waste disposal
- Miscellaneous

Natural sources

- Vegetation (organic compounds, bioaerosols)
- Volcanic eruption (CO, CO_2, NO_x, SO_x)
- Forest fires
- Biodegradation (H_2S, 2/3 of SO_x in atmosphere)
- Miscellaneous









Sources of air pollution : -- human activities

What pollutants are produced?

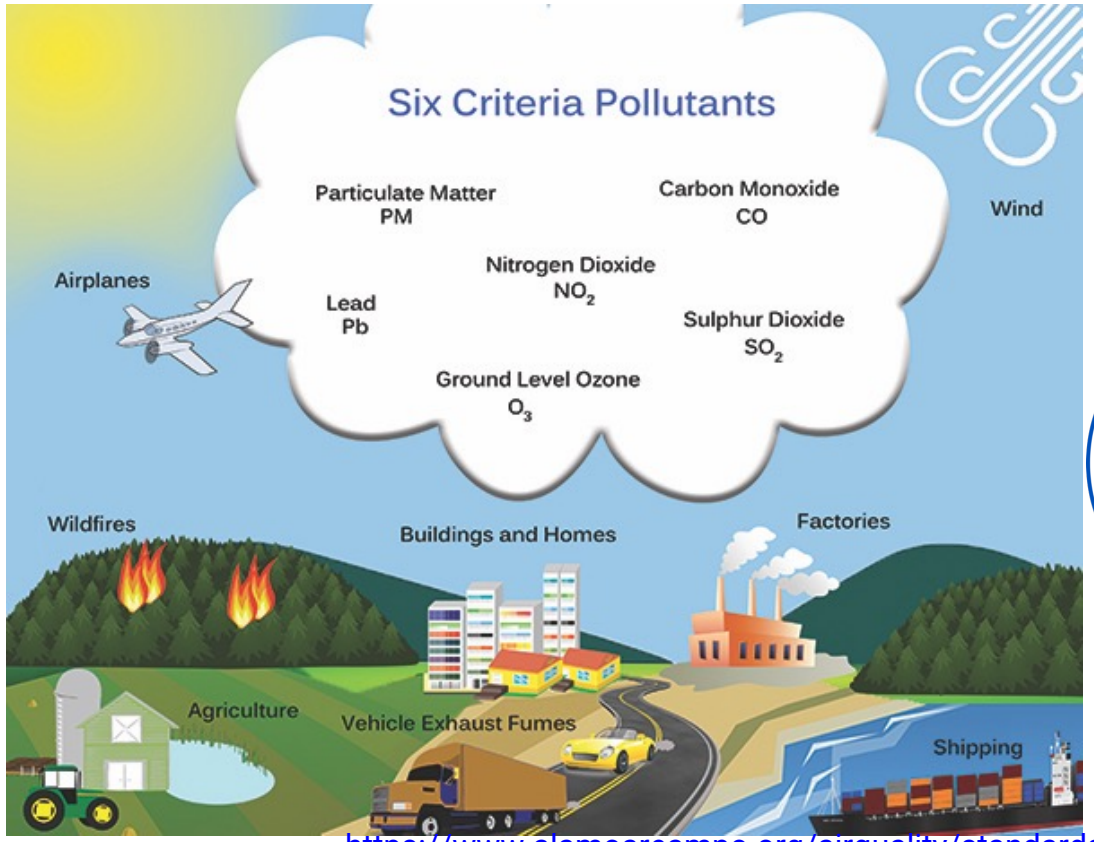
* ***Criteria pollutants***

Gases
- CO
- NOx
- SOx
- O_3

Particles
- PM2.5
- PM10

Toxic metals
- Pb

Reactive precursors
- VOCs
- NOx radicals



https://www.alamoareampo.org/airquality/standards.html

National Ambient Air Quality Standards

Pollutant		Primary/ Secondary	Averaging Time	Level
Carbon Monoxide (CO)		primary	8 hours	9 ppm
			1 hour	35 ppm
Lead (Pb)		primary and secondary (1)	Rolling 3 month average	0.15 µg/m^3
Nitrogen Dioxide (NO_2)		primary	1 hour	100 ppb
		primary and secondary	1 year	53 ppb (2)
Ozone (O_3)		primary and secondary (3)	8 hours	0.070 ppm
Particle Pollution (PM)	$PM_{2.5}$	primary	1 year	9.0 µg/m^3
		secondary	1 year	15.0 µg/m^3
		primary and secondary	24 hours	35 µg/m^3
	PM_{10}	primary and secondary	24 hours	150 µg/m^3
Sulfur Dioxide (SO_2)		primary (4)	1 hour	75 ppb
		secondary (5)	1 year	10 ppb

https://www.epa.gov/criteria-air-pollutants/naaqs-table

Criteria polluants - Carbon monoxide (CO)

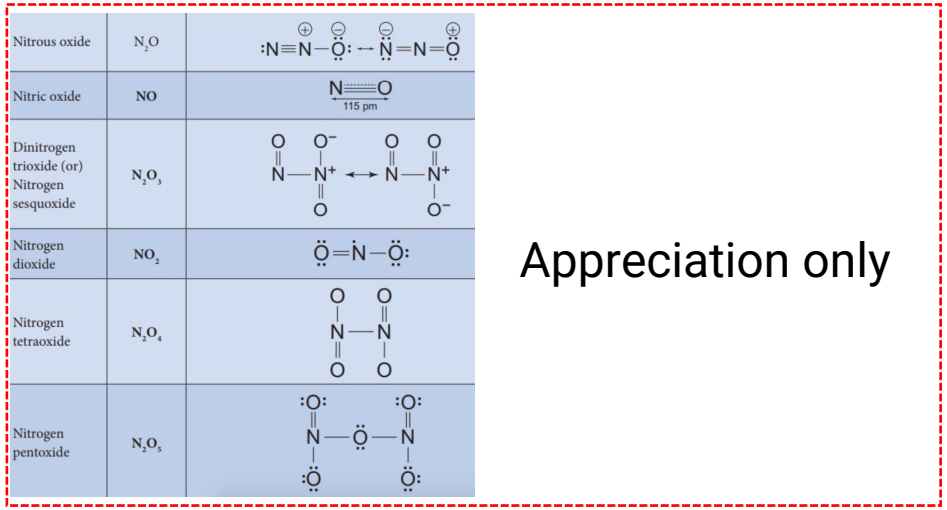
- Source
  - Incomplete combustion of fossil fuel.
    - Insufficient oxygen, low combustion temperature, insufficient combustion time, insufficient mixing
    - Combustion conditions better controlled in stationary sources → less CO emission
  - > 80% CO emission from transportation
- Health impacts
  - affect blood's ability to carry oxygen
    - Binds with hemoglobin to form carboxyhemoglobin (COHb)
    - Reduces oxygen carrying ability

Criteria polluants - Oxides of nitrogen (NOx)

Among all known nitrogen oxides, only two are important air pollutants

- Nitric oxide (NO)
- Nitrogen dioxide (NO_2)

} **NOx**



Appreciation only

Source

- **Thermal NOx** – formed by the reaction of N_2 and O_2 at high temperature (~ 1000 K)
- **Fuel NOx** – formed by the oxidation of nitrogen compounds bound in the fuel
  - E.g., nitrogen contained in coal

Criteria polluants - Oxides of nitrogen (NOx)

NO – colorless gas

- Precursor of ground level ozone and photochemical smog
- No known adverse health effects at typical atmospheric concentrations
- Prone to be oxidized to NO_2

NO_2 – reddish brown gas

- Respiratory damage at > 0.7 ppm
- → photochemical smog formation (ozone)

Other environmental effects of NOx:

acid rain

eutrophication

$$2NO + O_2 \rightarrow 2NO_2$$

$$3NO_2 + H_2O \rightarrow 2HNO_3 + NO$$



Criteria polluants - Sulfur oxides (SOx)

- SOx refers to
  - SO_2
  - SO_3 (small amount)
- Source
  - Combustion of S containing fossil fuel $S + O_2 \rightarrow SO_2$
    - Coal contains 1-6% S
    - Natural gas contains H_2S (hydrogen sulfide), which can be removed. Cleaned natural gas contains negligible amount of S.
  - Mainly from coal-fired power plants

$$SO_2 + \tfrac{1}{2} O_2 \rightarrow SO_3$$
$$SO_3 + H_2O \rightarrow H_2SO_4$$

Acid rain (acid deposition)

Criteria pollutants - Sulfur oxides (SOx)

- Effect
  - Health
    - Highly soluble in water → affect moist passage of the upper respiratory system
    - May combine with particulates to reach deeper into the lungs → greater damage
  - Damage vegetation
  - Corrosion of building materials

- Natural rainfall has a pH of 5 – 5.6
  - Controlled mainly by CO_2
- Acid rain - rainfall with lower pH
  - SOx
  - NOx

Natural rain ≈ 5.6
Acid rain ≈ 4.0–4.5
Lemon juice ≈ 2

$CO_2 + H_2O \rightarrow H_2CO_3$

$H_2CO_3 \rightarrow H^+ + HCO_3^-$





Similarities with SO_x

- Both SO_x and NO_x react with oxygen and water to form acids (sulphuric and nitric).

✓ Both are the principal precursor of acid rain.

✓ Both are primarily removed by rain.

- Both cause respiratory problems at high doses.
- Both are emitted largely by combustion.



Criteria polluants - Lead (Pb)

Source

- Mainly from leaded gasoline
- Emitted mainly in the form of inorganic particulates, settling down in vicinity of source (e.g., highways)
- Switch to non-leaded gasoline → greatly reduced lead emission
- Other sources: leaded paint, pipelines (leaded solder at joints)
- Exposure
  - Inhalation
  - Water
- Health effect
  - Reduced intelligence and detrimental effects on growth and development of children
  - Brain damage



Criteria pollutants - Ground-level ozone (O_3)

Source

- Secondary pollutant produced when NOx, VOCs, and sunlight come together
- The most abundant photochemical oxidant



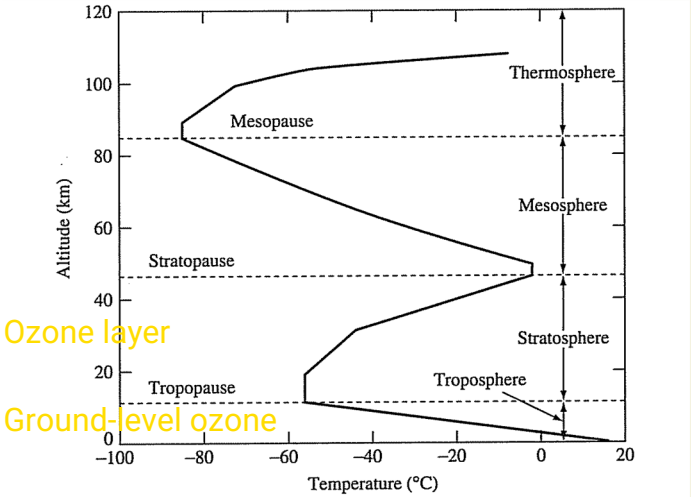


FIGURE 1 The U.S. Standard Atmosphere, showing the four major layers.

Effects

- Health: strong respiratory effects (e.g., coughing, shortness of breath, chest constriction, irritation of mucous membrane)
- Damage of vegetation
- Cracking of rubber

Photochemical reaction

$2\ NO + O_2 \rightarrow 2\ NO_2$

NO_2 + hv (photon) → NO + O (free atomic oxygen)

$O + O_2 \rightarrow O_3$

$O_3 + NO \rightarrow NO_2 + O_2$

} NO_2 promotes ozone formation

NO destroy ozone

Photochemical reaction - role of VOCs

$O + H_2O \rightarrow 2\ OH^{\bullet}$

$2\ OH^{\bullet} + NO + \text{organic compound} \rightarrow NO_2 + \text{oxidized organic compound (e.g., aldehyde)}$

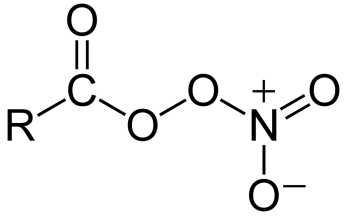
$$3CH_4 + 2O_3 \longrightarrow 3HCHO + 3H_2O$$

Formaldehyde

Other important components of photochemical smog

formaldehyde (HCHO)
peroxyacetyl nitrate (PAN)

Eye irritation



PAN

VOCs

Volatile organic compounds (VOCs) are volatile chemicals that contain organic carbons

- Examples: formaldehyde, benzene

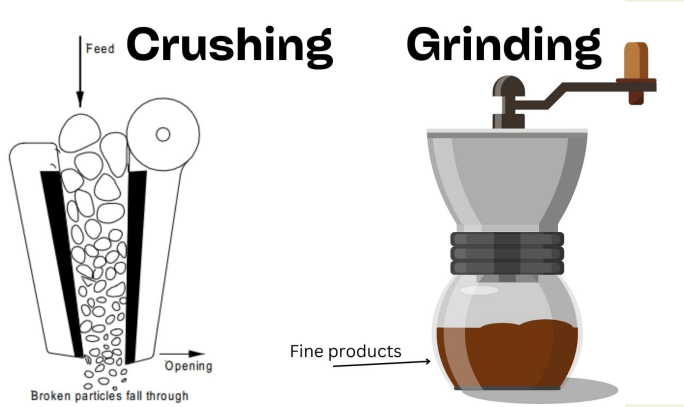
Source

- Evaporation of solvents, fuels, and other organics
- Partially burned fuels
- Effects
  - Precursors of photochemical smog
  - Health effect: risks associated with indoor exposure to some VOCs (see indoor air quality notes)

Criteria polluants - Particulate matter (PM)

Particulate matter *

- Nonhomogeneous pollutants
- Fine solids or liquids suspended in the atmosphere
- 0.005 – 100 μm in size
- Particulate matter is also termed aerosols
- Dusts – solid particles by grinding or crushing
- Mist, fog – liquid aerosols
- Smoke – particles composed primarily of carbon resulting form incomplete combustion




Grinding


Fog




Smoke

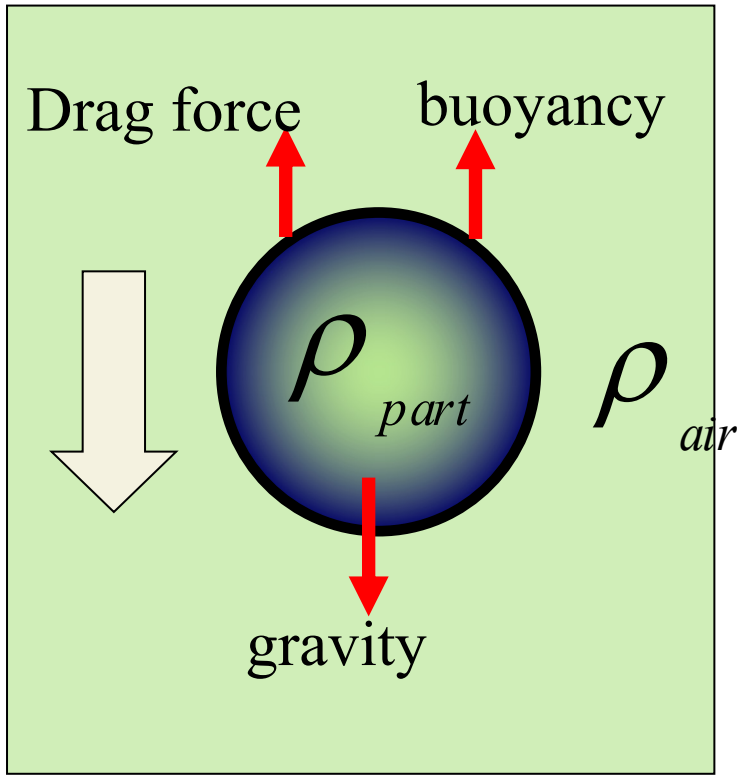
Terminal Settling Velocity

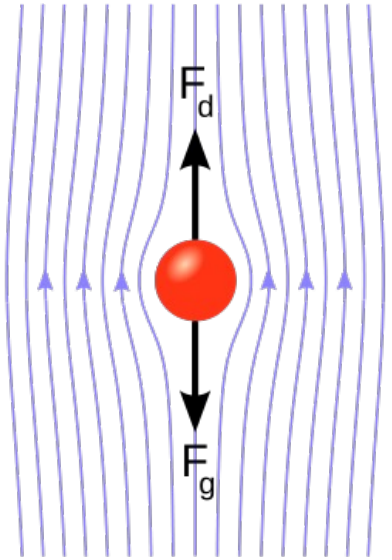
If $\rho_{particle} \gg \rho_{fluid}$

Terminal settling velocity

Stokes' Law

$$V_t = \frac{gD^2\rho_{part}}{18\mu}$$



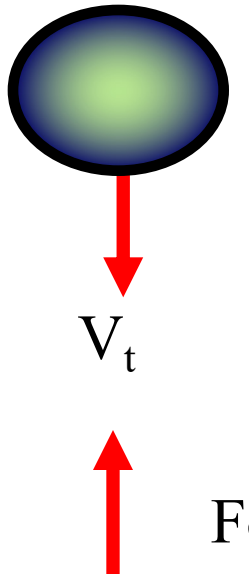


Units:
g = 9.81 m/s^2

D - m (radius of the spherical particle)
ρ - kg/m^3
μ - kg/m.s = 0.0172 g/m.s (dynamic viscosity)

Why are We Interested in Settling Velocity?

$$V_t = \frac{gD^2\rho_{part}}{18\mu} \quad \Rightarrow \quad V_t \propto D^2$$



1 μm $V_t = 0.006$ cm/s

10 μm $V_t = 0.6$ cm/s

1000 μm $V_t = 600$ cm/s

For u: upward wind velocity

For smaller particles, $u > V_t$ Suspending in the atmosphere for a long time.

For larger particles, $u < V_t$ Particles will settle to the ground, not causing air pollution.

Criteria polluants - Particulate matter (PM)

Important characteristics

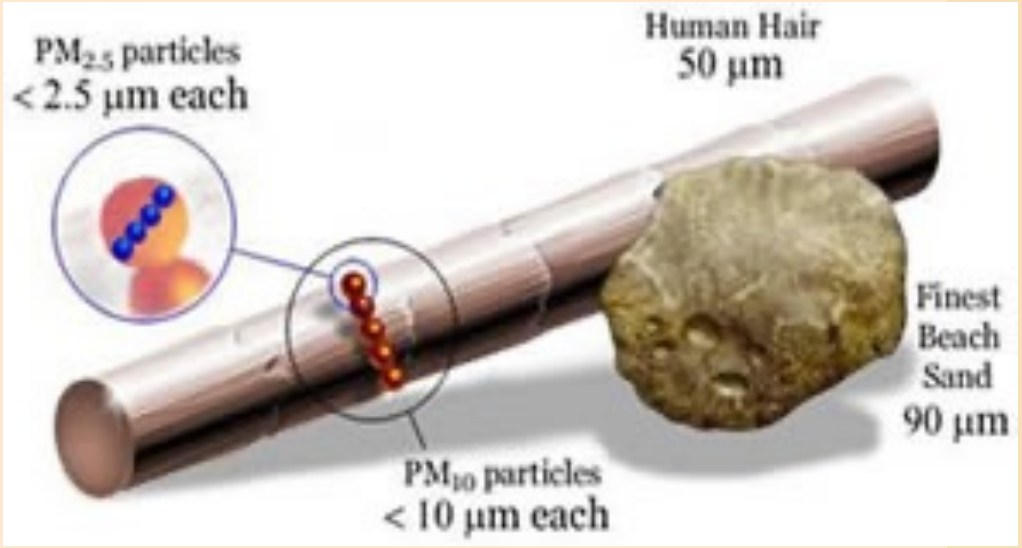
- Size, density, composition, phase (liquid, solid)

PM_{10} and $PM_{2.5}$

- PM_{10} – particulate matter less than 10 μm
- $PM_{2.5}$ – particulate matter less than 2.5 μm

Primary and secondary PM

- Primary PM – emitted directly into the atmosphere from the source and remaining unaltered
- Secondary PM – formed by primary PM reacting with gases or moisture in the atmosphere

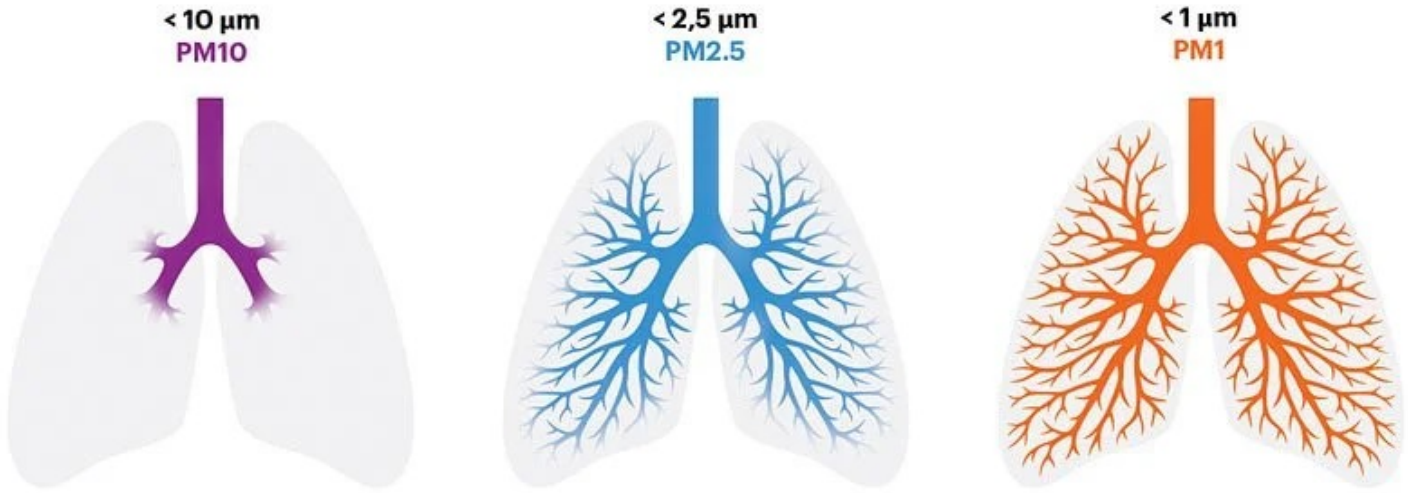


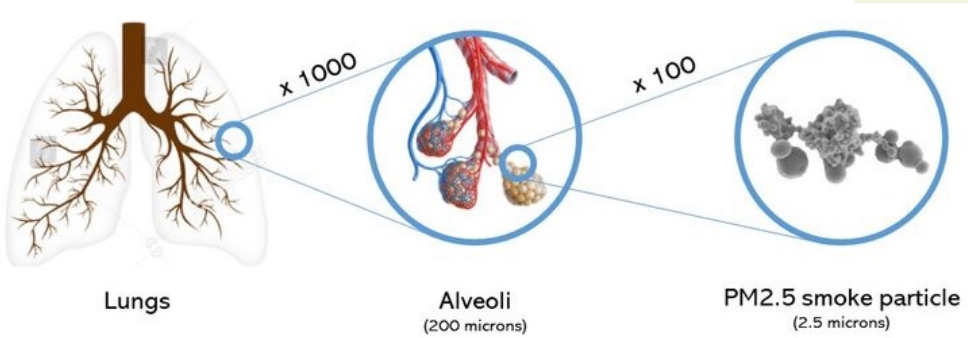
http://www.epa.gov/air/oaqps/eog/bces/module3

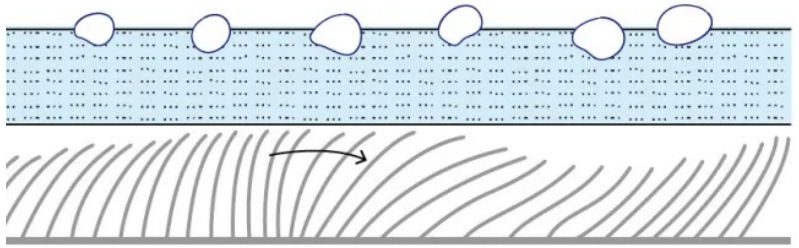
Ambient PM2.5 is the largest driver of air pollution's burden of disease worldwide, accounting for 62% of all global air pollution deaths (STATE OF GLOBAL AIR 2025).

Criteria polluants - Particulate matter (PM)*

- Particle size distribution are important in determining the behavior of particles in the respiratory tract (**health concern**).
- PM2.5 are more dangerous. They get deeper into the lungs. Reach all parts of the lungs and do not leave the lungs with mucus like larger particles







[Motor vehicle pollutants]

nasal hair and mucus

Example

NOx is one of the main air pollutants, please

(1) List at least two sources from which these pollutants originate?

(2) List at least two suggestions on how to control these pollutants?

Solutions: Open-ended questions, potential key points include:

(1) Fossil fuel combustion; Thermal NOx; Use of nitrogen-based fertilizers (nitrification, soil microbes convert ammonia (NH_3) into nitrate (NO_3^-) and release NO and NO_2); Volcanic activities, et al.

(2) Cleaner fuels, vehicle emission controls, public transportation, et al.

How do we measure air pollution? Air quality index (AQI)

A scale for air quality, devised by the U.S. Environmental Protection Agency (USEPA)

Followed by many countries/regions

- Singapore, China, Thailand, Malaysia, South Korea, … (with some modifications)

The index is based on measurements of the concentrations of five pollutants:

- particulate matter (**$PM_{2.5}$** / PM_{10})
- sulfur dioxide (SO_2)
- carbon monoxide (CO)
- nitrogen dioxide (NO_2)
- ozone (O_3)

Air quality index (AQI)

Air quality standards have been developed for each of the five pollutants.

After determining a value for each of the five pollutants, the highest of the five numbers is reported as the (Pollutant Standards Index) PSI for the day.

Index	Designation	1 hr O_3 (ppm)	8 hr CO (ppm)	24 hr PM 10 ($\mu g/m^3$)	24 hr SO_2 (ppm)	1 hr NO_2 (ppm)
0	—	0	0	0	0	—[a]
50	—	0.06	4.5	50	0.03	—[a]
100	NAAQS	0.12[b]	9	150	0.14	—[a]
200	Alert	0.20	15	350	0.30	0.6
300	Warning	0.40	30	420	0.60	1.2
400	Emergency	0.50	40	500	0.80	1.6
500	Significant harm	0.60	50	600	1.00	2.0

[a]No index values reported at concentrations below the Alert level.
[b]does not yet reflect 1997 change in standard.
Source EPA, 1994b.

Example for AQI calculation

0.16 ppm O_3 -> value for O_3 = 150
5.4 ppm CO -> value for CO = 60
350 µg/m^3 PM_{10} -> value for PM_{10} = 200
0.14 ppm SO_2 -> value for SO_2 = 100
0.72 ppm NO_2 -> value for NO_2 = 220

Maximum value is 220 (for NO_2)
⇒ PSI = 220 based on NO_2
⇒ very unhealthful air quality

Index	Designation	1 hr O_3 (ppm)	8 hr CO (ppm)	24 hr PM 10 ($\mu g/m^3$)	24 hr SO_2 (ppm)	1 hr NO_2 (ppm)
0	—	0	0	0	0	—[a]
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airnow.gov/aqi/aqi-calculator-concentration/

AirNow | AirNow | AQI & Health | Fires | Maps & Data | Education | International | Resources | Español Beta

Get Current and Forecast Air Quality for Your Area | ZIP Code, City, or State

AQI Calculator

AQI Calculators

- AQI to Concentration
- Concentration to AQI

Concentration to AQI Calculator

DIRECTIONS: *Select a pollutant, then enter the Concentration value. Click on "Calculate" to see the results.*

Select a Pollutant

PM2.5 - Particulate <2.5 microns (24hr avg)

Units

ug/m3

Enter the Concentration

34

Calculate Reset

AQI

97

AQI Category

Moderate

Sensitive Groups

People with respiratory or heart disease, the elderly and children are the groups most at risk.

Air quality index (AQI)

Appreciation only

Good (0-50): no health impacts

Moderate (51-100): unusually sensitive people should consider reducing prolonged or heavy exertion

Unhealthy for sensitive groups (101-150): active children and adults, and people with lung disease should reduce prolonged or heavy exertion outdoors. The general public is not likely to be affected.

Unhealthy (151-200): everyone may begin to experience health effects. Members of sensitive groups may experience more serious health effects.

Very unhealthy (201-300): air quality in this range triggers a health alert, meaning every one may experience more serious health effects.

Hazardous (301-500): Air quality in this range triggers health warnings of emergency conditions. The entire population is more likely to be affected.

Air Quality Health Index (AQHI)

Hong Kong switched from AQI to AQHI from 30 December 2013

The AQHI informs the public of the short-term health risk of air pollution

The AQHIs are reported on a scale of 1 to 10 and 10+ and are grouped into five health risk categories:

Health Risk	Low			Moderate			High	Very High			Serious
AQHI Scale	1	2	3	4	5	6	7	8	9	10	10+

http://www.aqhi.gov.hk/en/what-is-aqhi/about-aqhi.html

Air Quality Health Index (AQHI)

AQHIs are calculated based on the cumulative health risk attributable to the 3-hour moving average concentrations of

- Ground-level ozone (O_3)
- Nitrogen dioxide (NO_2)
- Sulphur dioxide (SO_2)
- Particulate matter ($PM_{2.5}$ / PM_{10})

The risk factors of each pollutant were obtained from local health studies

Combined health effects

More details can be found in http://www.epd.gov.hk/epd/english/environmentinhk/air/studyrpts/files/APIreview_report.pdf

http://www.aqhi.gov.hk/en/what-is-aqhi/about-aqhi.html

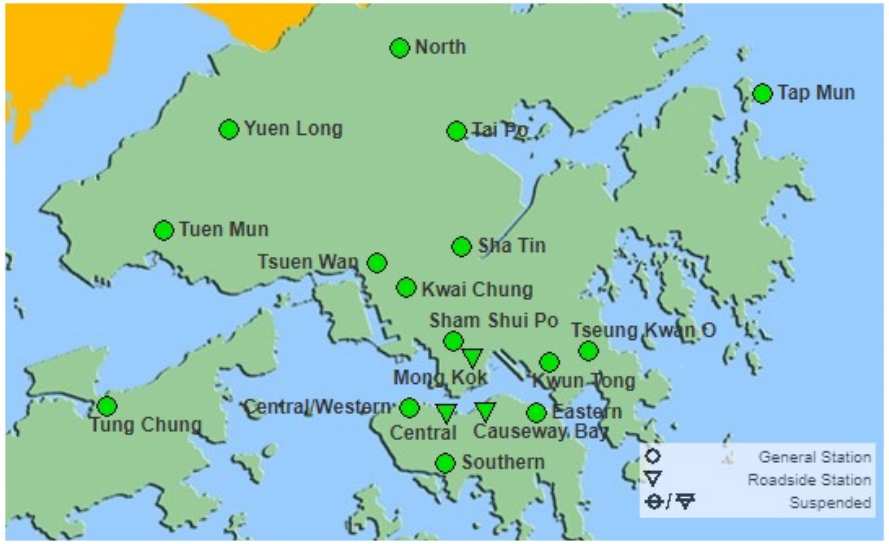
Air Quality Health Index (AQHI)

[EPD - AQHI and Forecast]

Current AQHI [11:00 2024-02-19]

General Stations: 2 to 3 (Low)

Roadside Stations: 2 to 3 (Low)



Remarks:

(1) Time shown is in Hong Kong time.

(2) The AQHI information is based on real time data taken directly from EPD's Air Quality Monitoring Network.

(3) The hourly reported AQHI is for short term health risk communication; for health risks of long-term exposure of the air quality, please refer to the Annual Air Quality Index(Annual AQI).

(4) In case of equipment or station suspension due to repair and maintenance, the data collection for calculation of AQHI at station will be affected, the data of similar stations will then be adopted. Such AQHI will be shown with an asterisk.

FORECAST of Health Risk

19-02-2024	Today P.M.	Tomorrow A.M.
General Stations	Low	Low
Roadside Stations	Low	Low



Air Quality Health Index

11:00 19-02-2024		AQHI	Health Risk
General Stations	Central/Western	2	Low
	Southern	2	Low
	Eastern	3	Low
	Kwun Tong	2	Low
	Sham Shui Po	2	Low
	Kwai Chung	2	Low
	Tsuen Wan	2	Low
	Tseung Kwan O	2	Low
	Yuen Long	2	Low
	Tuen Mun	2	Low
	Tung Chung	2	Low
	Tai Po	2	Low
	Sha Tin	2	Low
	North	2	Low
	Tap Mun	2	Low
Roadside Stations	Causeway Bay	3	Low
	Central	2*	Low
	Mong Kok	3	Low

How do we control air pollution?

Mobile sources: Motor vehicle pollutants

CO – vehicle emission accounts for 75% of total CO emission.

NO_x - vehicle emission accounts for 40-50% of total NO_x emission

Hydrocarbon (HC) - vehicle emission accounts for 40-50% of total HC emission

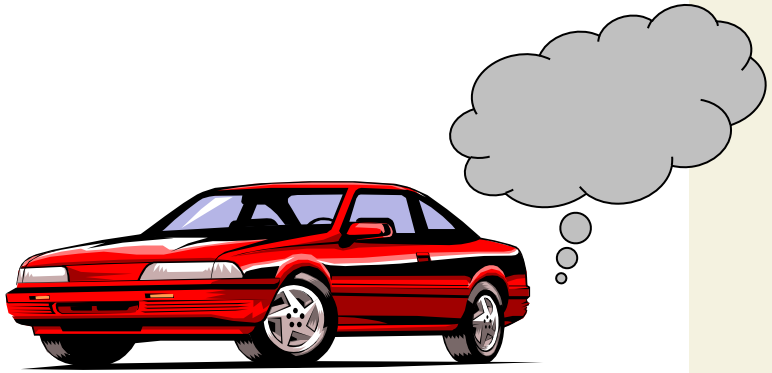
Lesser amounts of particulates and SO_x

Lead (Pb) – if leaded gasoline is used



Control measures

Reducing Vehicle Emissions

Emissions =  × 

Vehicle miles driven per day

Emissions per vehicle mile

Emission Factors

Emission Factors for Motor Vehicles (in g/mile)

Year	*CO*	*NO_x*	*HC*
Before 1960	87	3.6	8.8
1970	23	3.6	2.2
1972	39	3.6	3.4*
1975	15	3.1	1.5
1978	15	2	1.5
1980	7	2	0.41
1981	3.4	1	0.41
1993	3.4	0.4	0.25
2003	1.7	0.2	0.0125

* Change of testing procedure

Example 1:

Estimate how much CO, HC and NO_x are removed from atmosphere per year by removing an old (pre 1960) motorcar from the road and replacing it with a new (2003) model. Assume the vehicles travel an average of 15,000 miles per year.

Solution:

Emission factors for pre 1960 motorcars:

CO = 87 g/mile; NO_x = 3.6 g/mile ; HC = 8.8 g/mile

Emission factors for 2003 motorcars:

CO = 1.7 g/mile; NO_x = 0.2 g/mile ; HC = 0.0125 g/mile

CO emission reduction: (87-1.7) × 15000 = 1,279.5 kg/yr

NO_x emission reduction: (3.6-0.2) × 15000 = 51 kg/yr

HC emission reduction: (8.8-0.0125) × 15000 = 132 kg/yr

Motor Vehicle Emission Control



➢Looking for alternative fuels and power generation

- Ethanol; methanol; compressed natural gas (CNG); H_2
- Hybrid electric vehicles (HEVs); plug-in hybrid electric vehicles (PHEVs); Electric vehicles (EVs); fuel cell vehicles

➢Transportation control

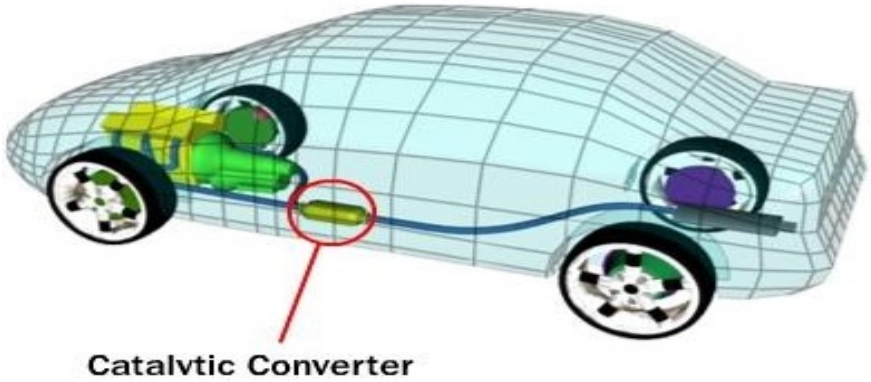
- Public transportation

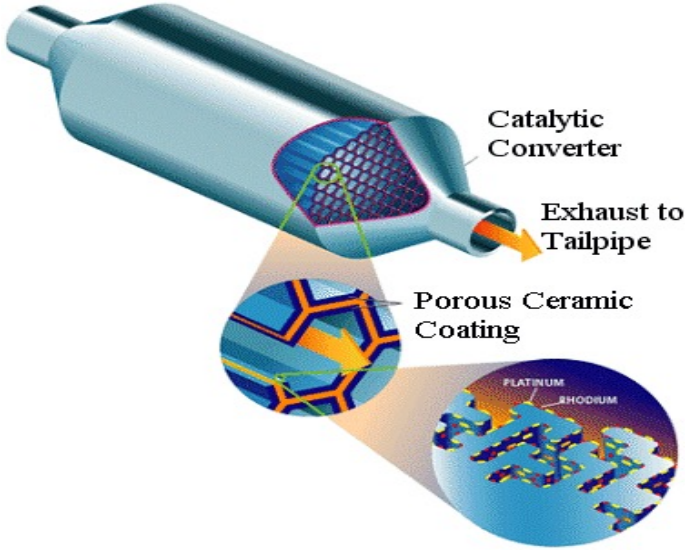
➢Add-on control

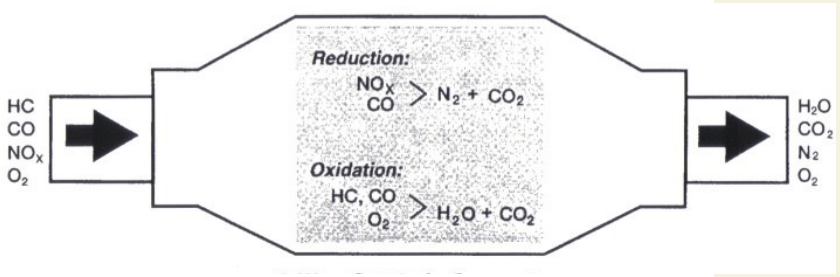
- Catalytic converter

Stationary sources

Catalytic converter







Catalysts: Platinum, Palladium, rhodium

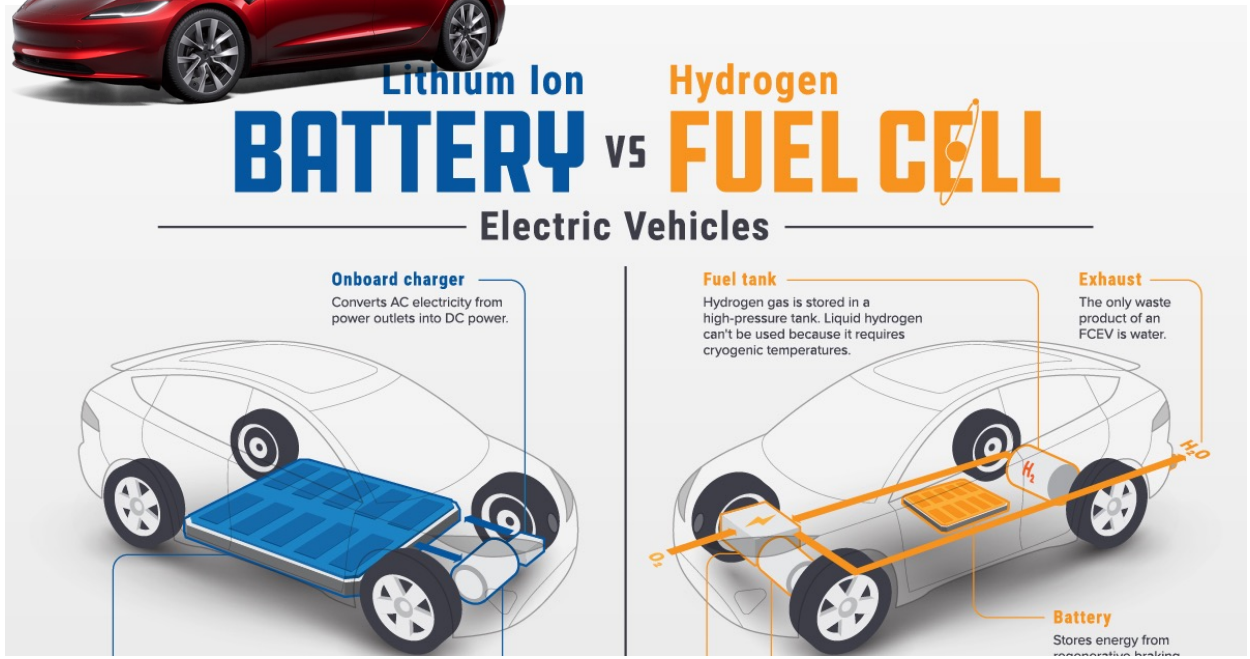
$$NOx + CO + HC \xrightarrow[\text{catalyst}]{\text{Pt-Pd-Rd}} N_2 + CO_2 + H_2O$$

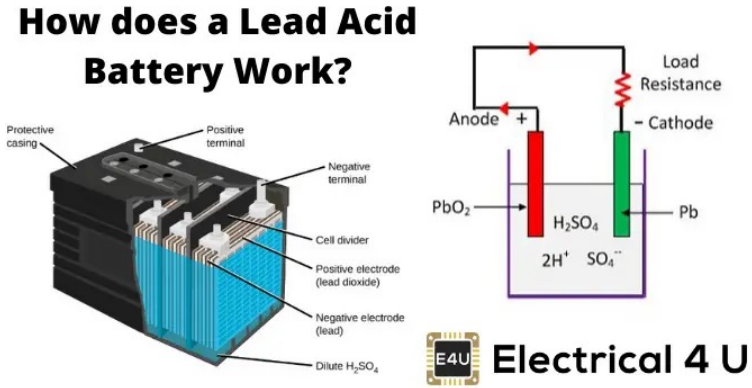
Exam question:
Which of the following is not production of catalytic converter? (a) CO_2 (b) H_2O (c) N_2 (d) SO_2, or (f) none of the above.

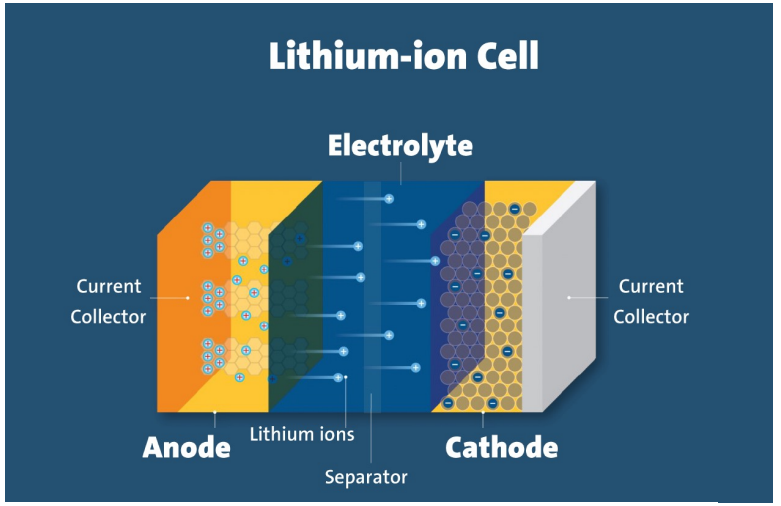
By 2001 275 million of the world's 500 million cars were fitted with catalysts. 85% of all new cars worldwide are fitted with catalytic converters.

Changes in Vehicle Systems









Reduction takes place at the cathode. There, cobalt oxide combines with lithium ions to form lithium-cobalt oxide ($LiCoO_2$). The half-reaction is:

$CoO_2 + Li^+ + e^- \rightarrow LiCoO_2$

Oxidation takes place at the anode. There, the graphite intercalation compound LiC_6 forms graphite (C_6) and lithium ions. The half-reaction is:

$LiC_6 \rightarrow C_6 + Li^+ + e^-$

Here is the full reaction (left to right = discharging, right to left = charging):

$LiC_6 + CoO_2 \rightleftarrows C_6 + LiCoO_2$

Stationary sources

Power plants (especially coal fired power plants) contribute most of SOx and significant portion of NOx and particulates

- Increase conversion efficiency from fuel to electricity
- Increase efficiency to energy use
- Use cleaner fuel



Control measures

- Pre-combustion: reduce emission potential of fuel (fuels with less S and N)
- Combustion control: improving combustion itself
- Post-combustion control: remove pollutants from stack gas

[Post-combustion control]

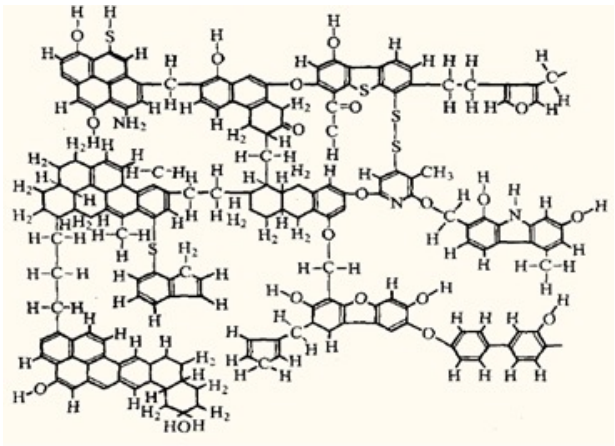
Pre-combustion control

Fuel switching

- Use lower S content coal
- Power plant coal S content: 0.2-5.5% → 0.17%
  - Low S coal: coal that emit less than 0.5 g/MJ

Coal cleaning

- S that is not chemically bond to coal molecules (e.g., pyrite (ferrous disulfide) FeS_2) → physical cleaning, magnetic separation
- Chemically bond S → biological/chemical treatment (at research stage)



Coal structures

Appreciation only

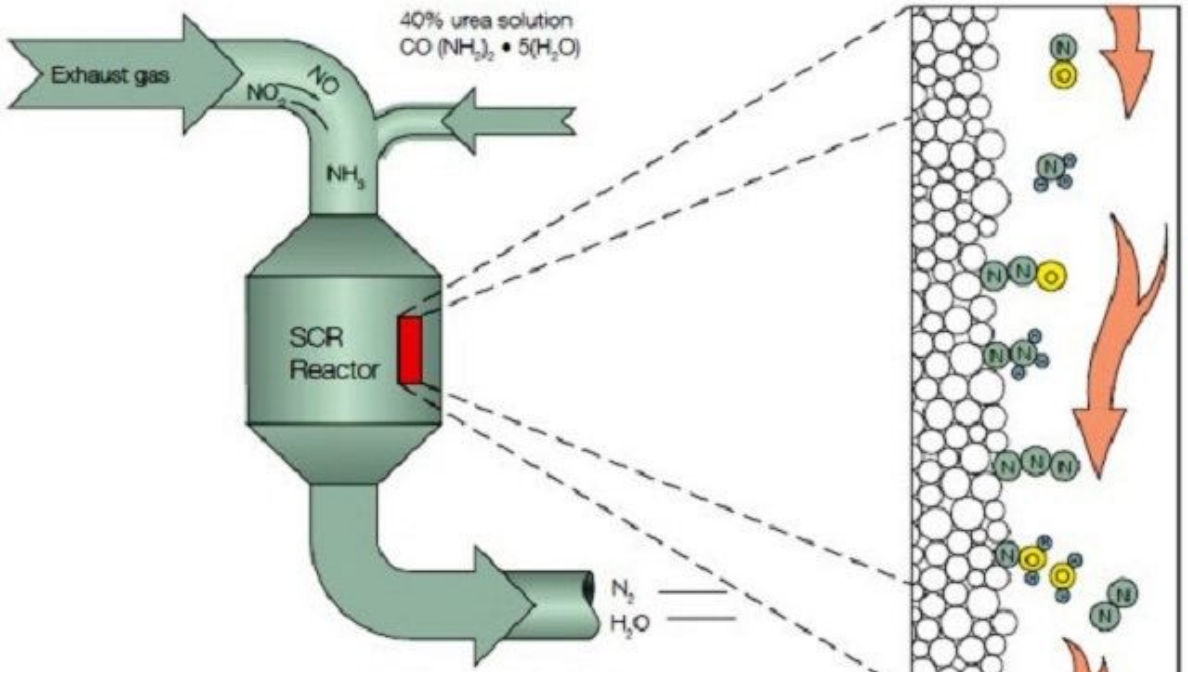
Post-combustion control - NOx emission

Selective catalytic reduction

Ammonia injected to boiler flue gas to reduce NOx

$$4NO + 4NH_3 + O_2 \rightarrow 4N_2 + 6H_2O$$

$$2NO_2 + 4NH_3 + O_2 \rightarrow 3N_2 + 6H_2O$$



Post-combustion control - SOx emission

Gas scrubber

- In a wet scrubber, limestone slurry sprayed into flue gas
- $CaCO_3 + SO_2 \rightarrow CaSO_3 + CO_2$
- $CaSO_3$ (calcium sulfite) removed as sludge

- Lime (CaO) can be used instead of limestone
  - Better removal efficiency but more expensive
- Dry scrubber (spray lime powders) is more expensive and less used

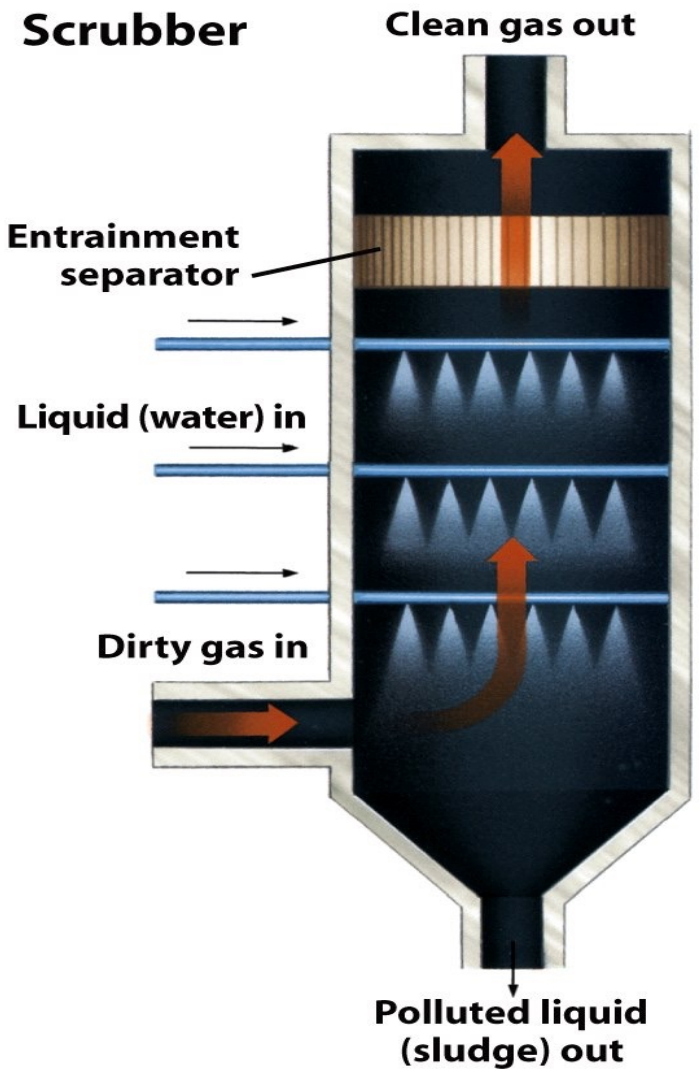
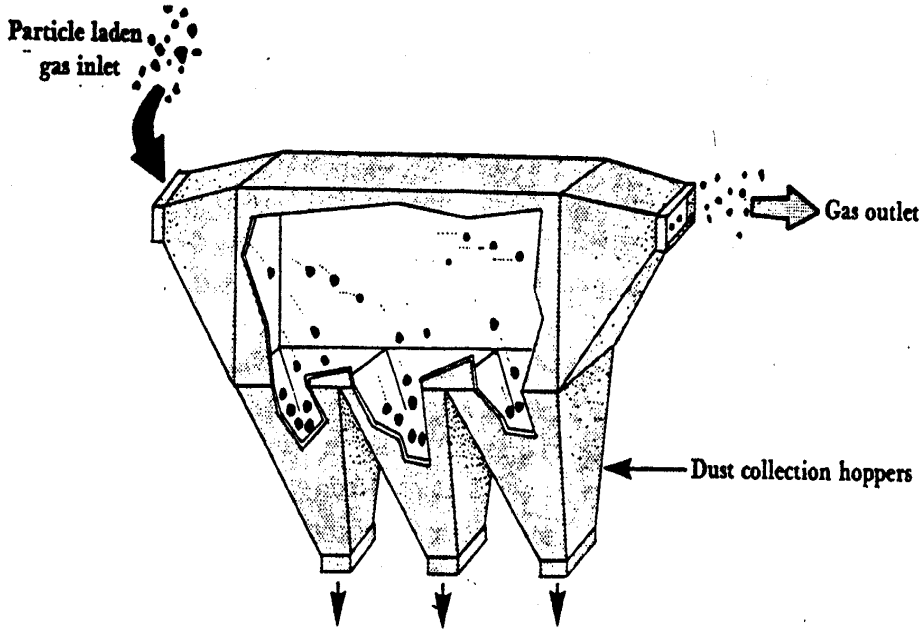
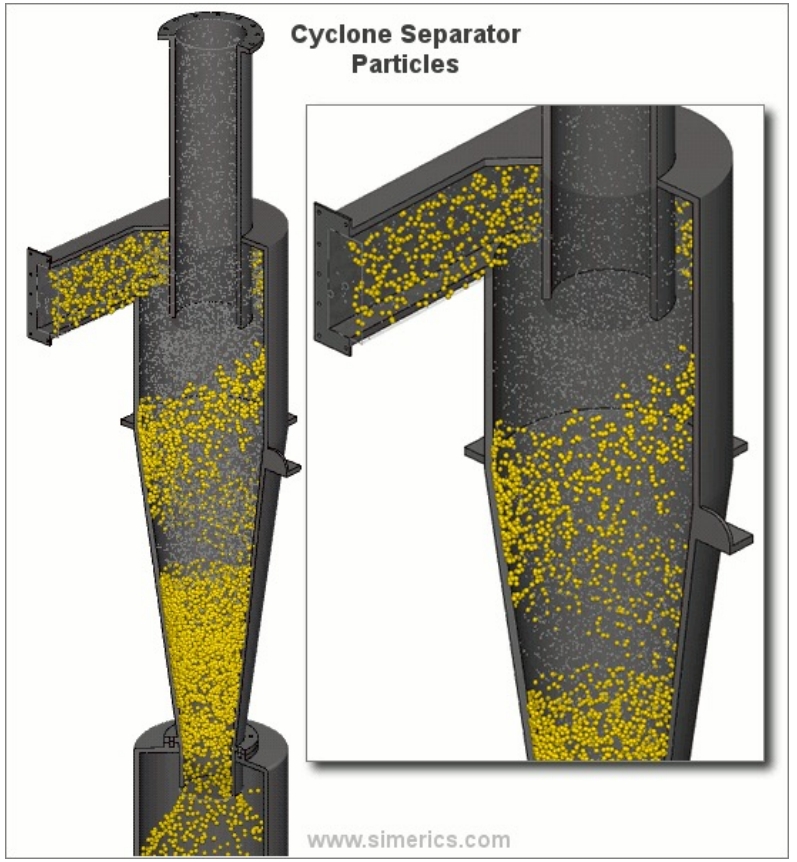


Figure 20-10d Environment, 5/e
© 2006 John Wiley & Sons

Post-combustion control - particulate removal



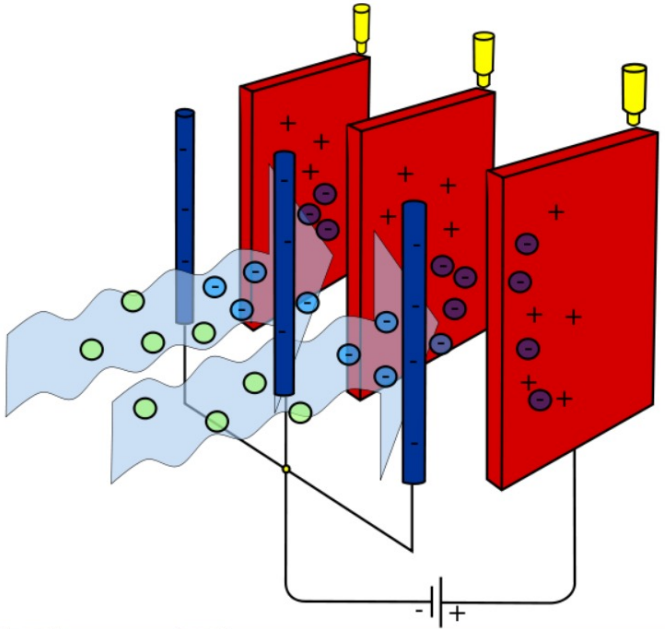
Gravity Settlers



Cyclones

- Reply on centrifugal force
- > 90% removal for particles > 5 μm
- Lower efficiency for smaller particulates

Post-combustion control - particulate removal



Electrostatic precipitators (ESP)

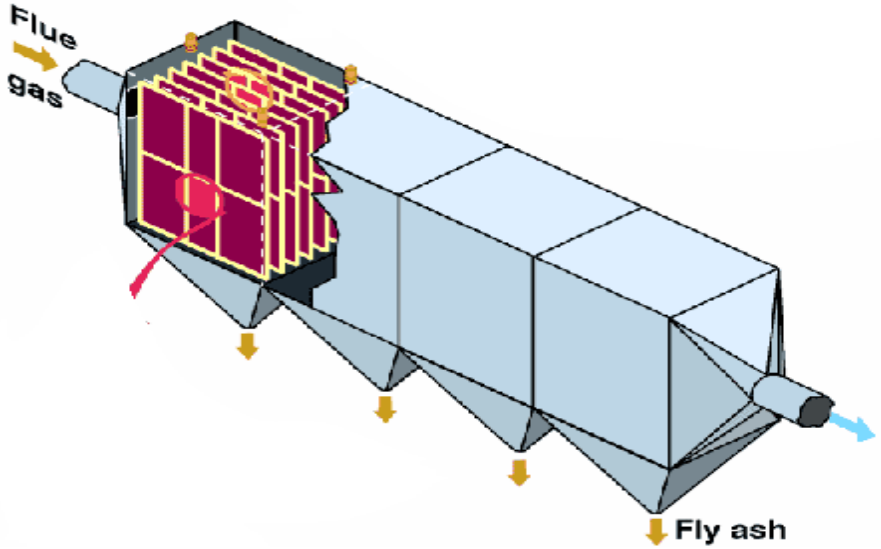
$$\eta = 1 - e^{-\omega A/Q}$$

Drift velocity

Area

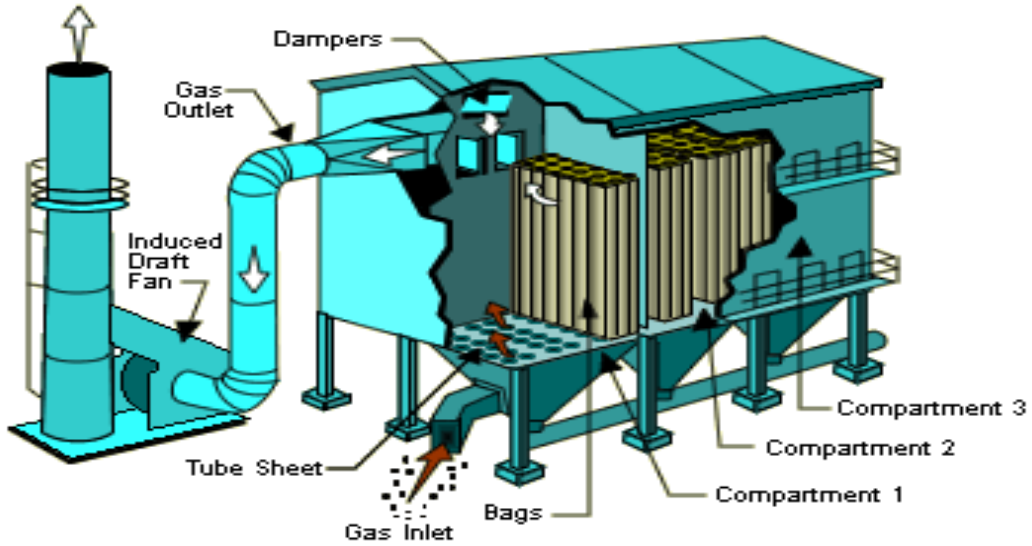
Flow rate

- Plates grounded, wires charged up to 100,000 V
- Gas molecules ionized
- Some negatively charged ions/electron attach to particles
- Charged particles collected by plates
- Removal efficiency (η)



Post-combustion control - particulate removal





Surface filter (Baghouse)

- Flue gas passing through fabric filters
- Near complete removal of > 1 μm particles
- Substantial removal of 0.01 μm particles



http://www.americanwoodworker.com/cfs-filesystemfile.ashx/__key/CommunityServer.Blogs.Components.WeblogFiles/tools/Baghouse_5F00_Bags_5F00_JPG.JPG

Example

1. In post-combustion control of NOx emission, selective catalytic reduction (SCR) is widely applied. Which of the following is used to reduce NOx? A, N_2O B, NH_3 C, N_2 D, $CaCO_3$

2. To remove SOx in post-combustion control, Which of the following is NOT used? A, CaO B, $CaCO_3$ C, Lime powders D, $CaSO_3$

Air pollution and meteorology

Stable air:

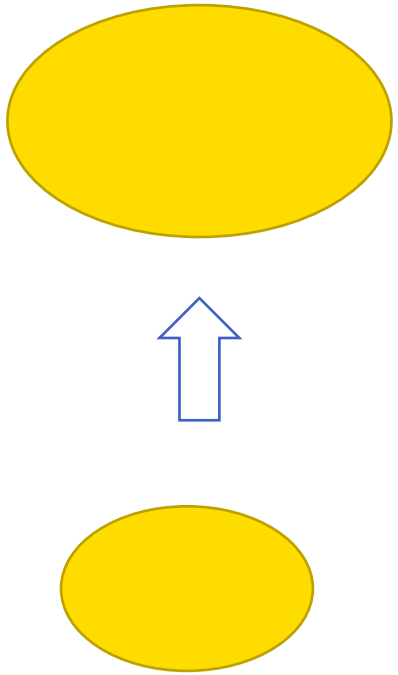
- physical forces acting on air such as it wants to remain at the given elevation
- reduced vertical dispersion and dilution of pollutants
- vertical stability of air depends on the temperature profile



A temperature inversion led to a stable air and little dispersion of air pollutants at Four Corners power plant.

Source: Davis and Masten, 2009, Principles of Environmental Engineering and Science (2nd Edition).

Adiabatic lapse rate



When a parcel of air moves upward in atmosphere

- Pressure reduces → volume expand (doing work to the surrounding)
- → Temperature reduces
- Assuming no heat exchange with the surrounding air (adiabatic condition):
- dT/dZ = - 10 °C/km (for dry air, Γ_{dry})
  - → temperature of the air parcel decreases by 1 °C for every 100 m increase in height
- For saturated air, reducing temperature → condensation and release to latent heat → temperature does not decrease as fast as Γ_{dry}

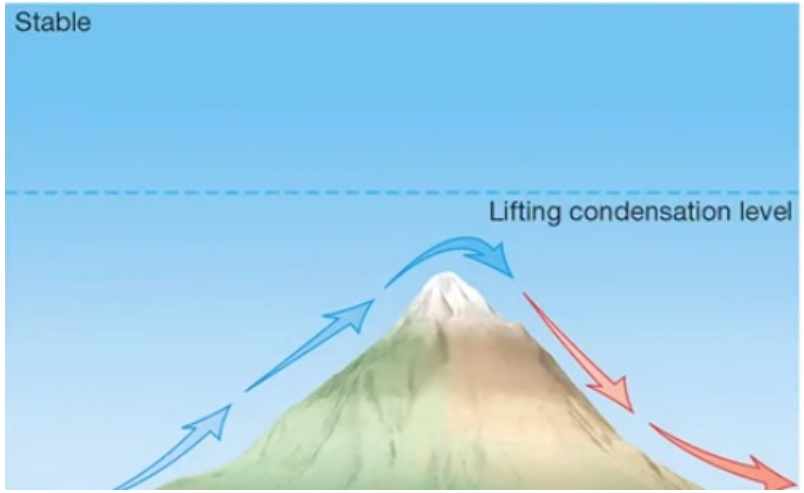
  ($\Gamma_{saturation}$ = - 6 °C/km)

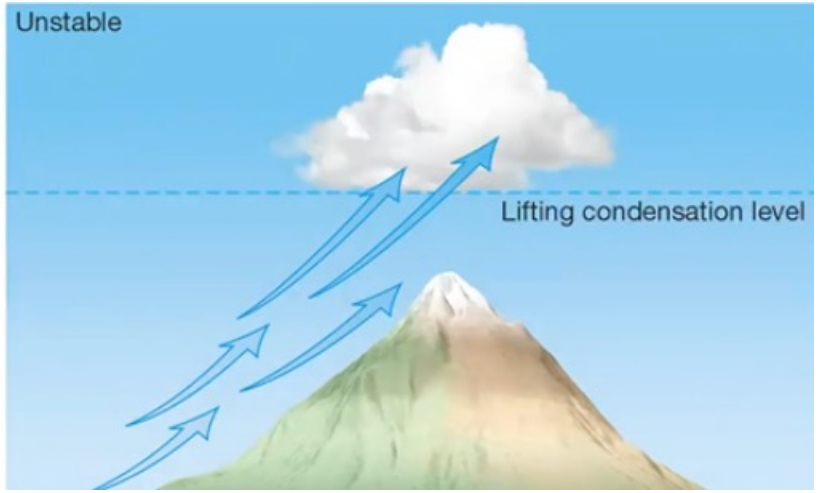
Adiabatic lapse rate

The actual temperature profile may differ from the adiabatic lapse rate

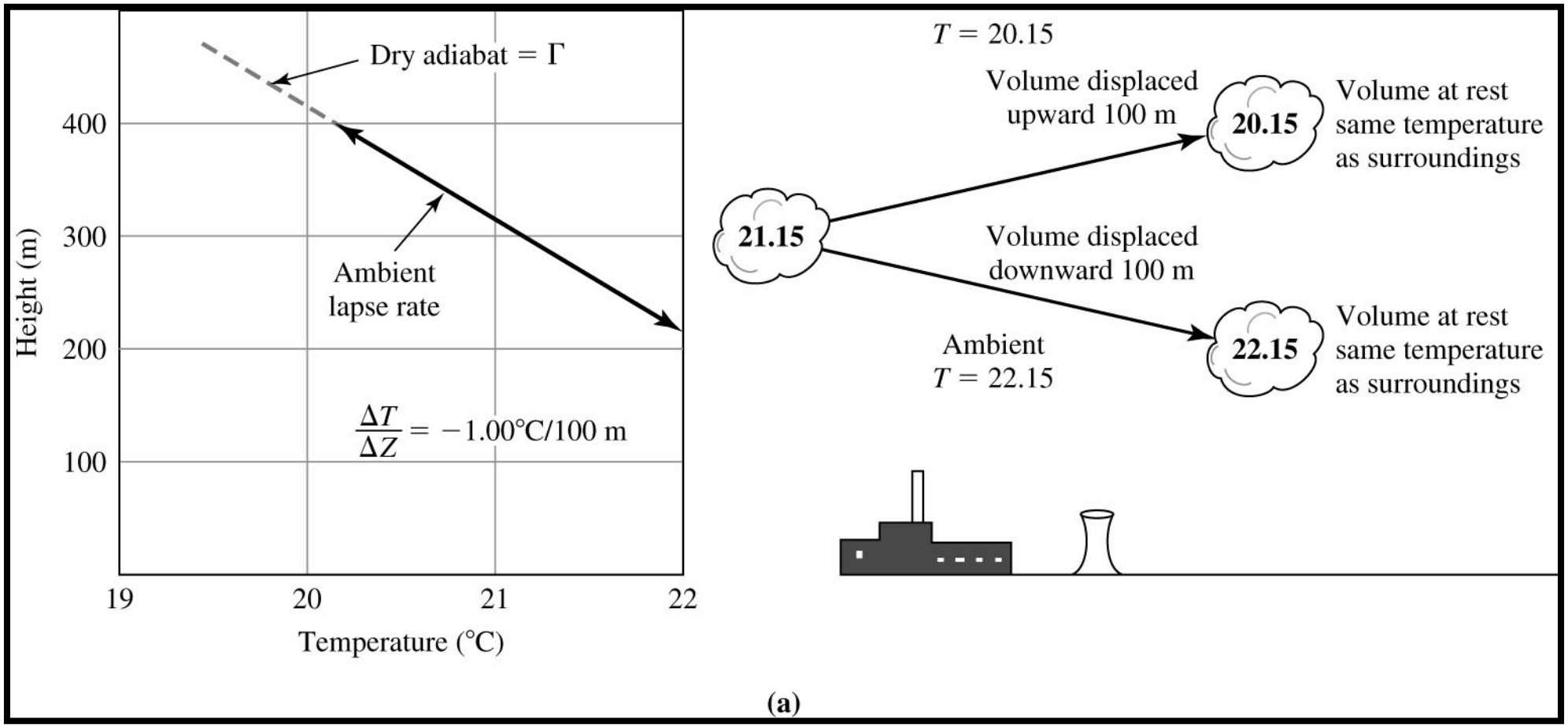
- Wind
- Sunlight
- Geographical features
- …

Ambient lapse rate refers to the rate at which the actual atmospheric temperature changes with elevation



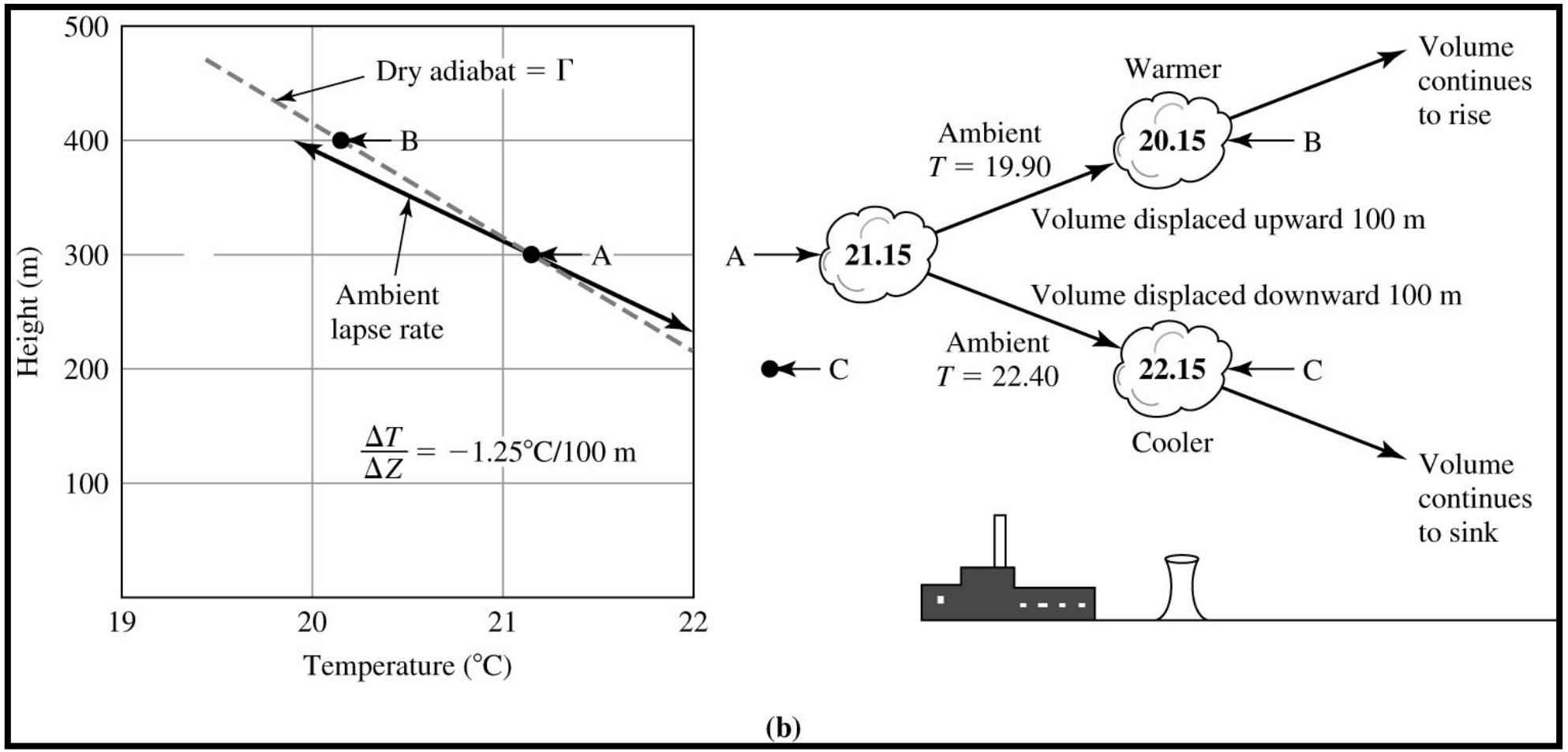


Atmospheric stability – neutrally stable



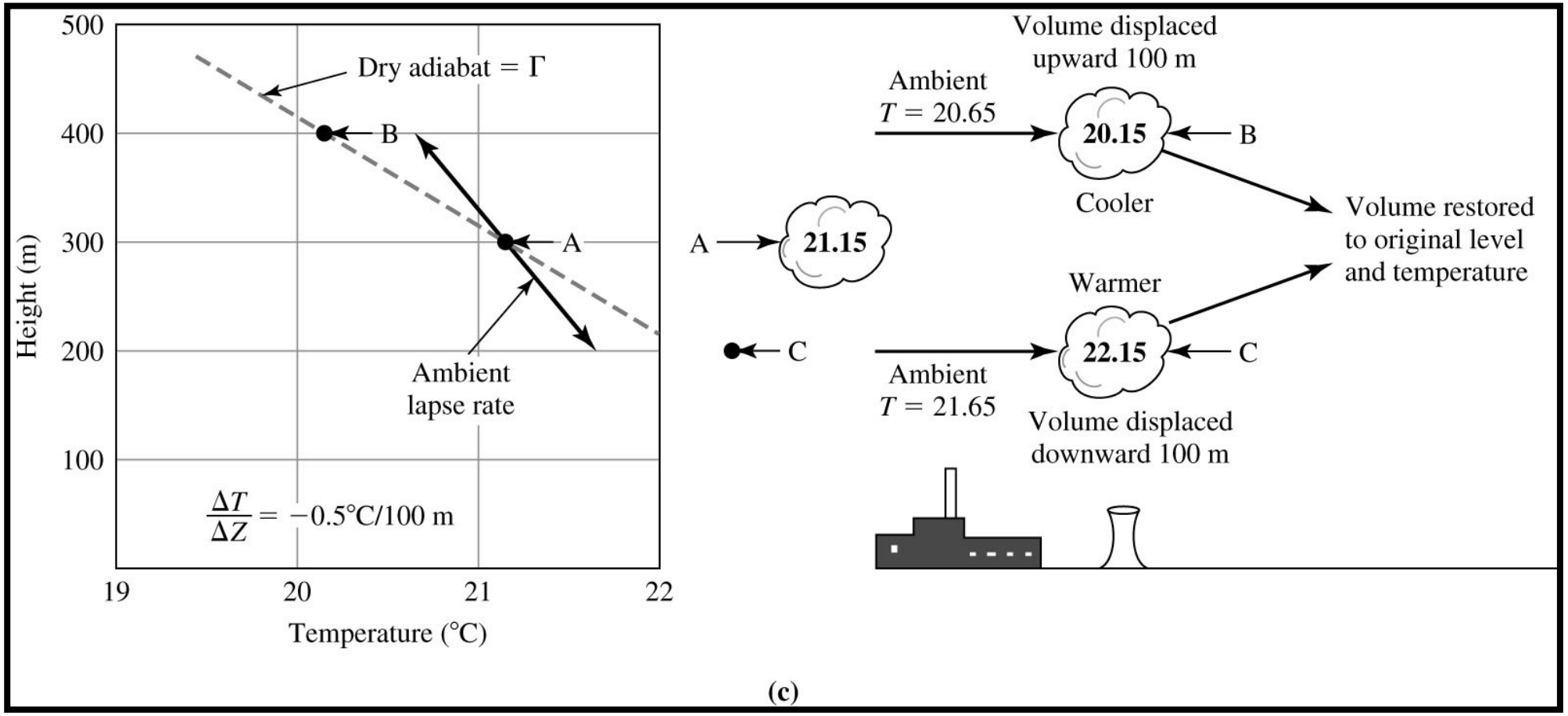
Source: Davis and Masten, 2009, Principles of Environmental Engineering and Science (2nd Edition).

Atmospheric stability – unstable



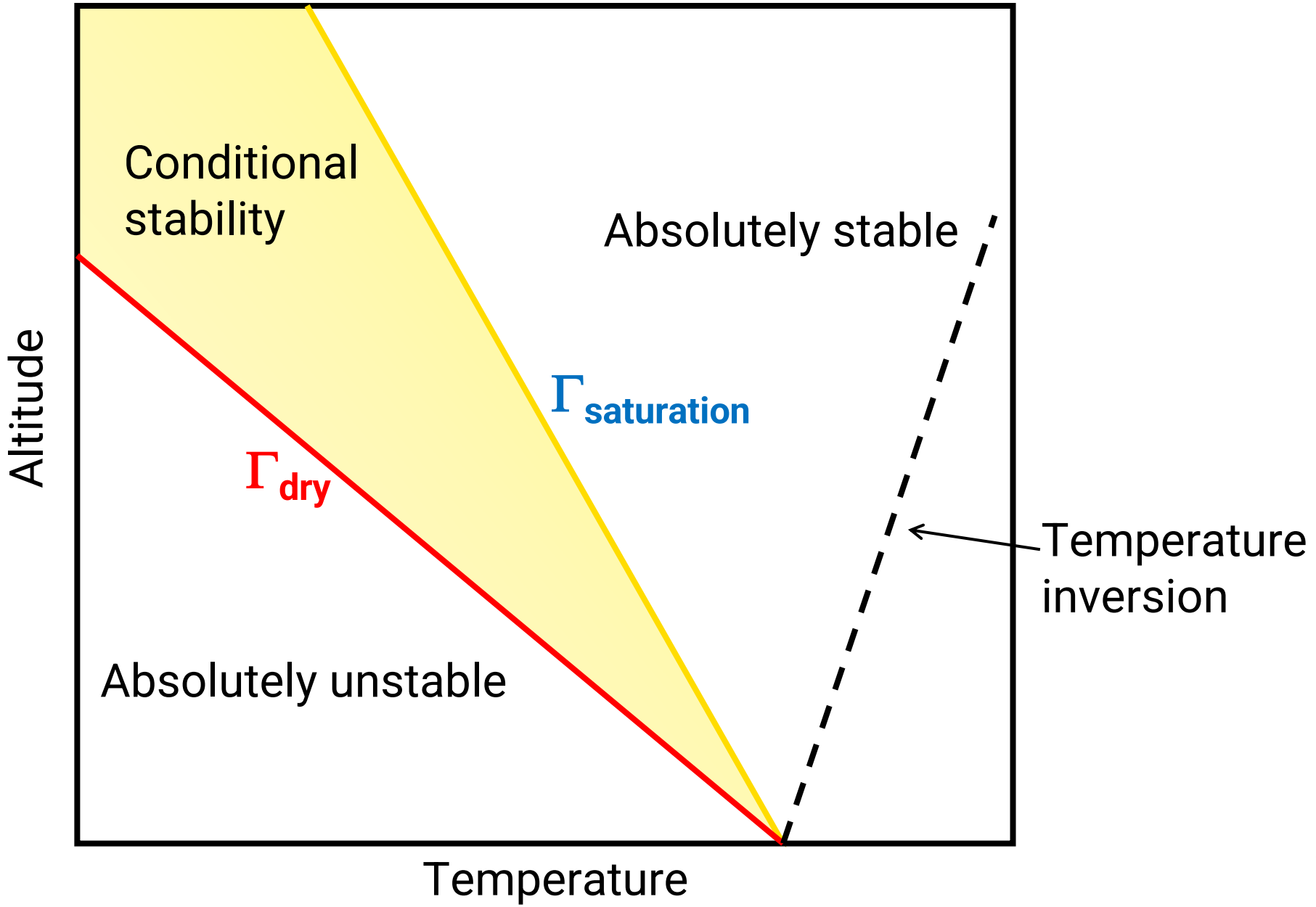
Source: Davis and Masten, 2009, Principles of Environmental Engineering and Science (2nd Edition).

Atmospheric stability – stable



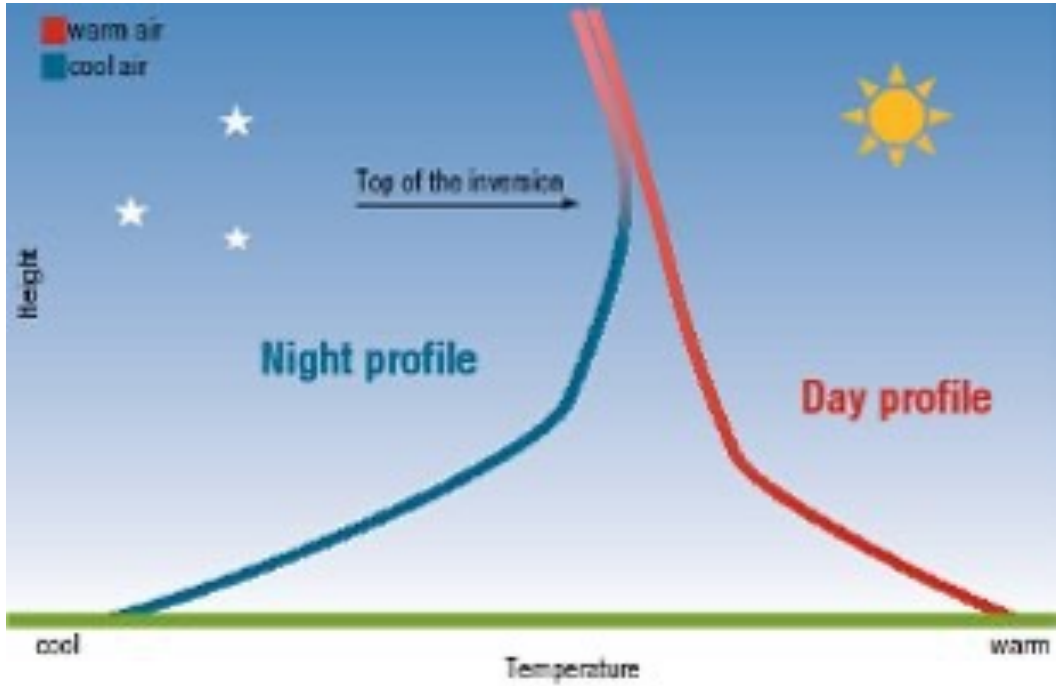
Source: Davis and Masten, 2009, Principles of Environmental Engineering and Science (2nd Edition).

Atmospheric stability

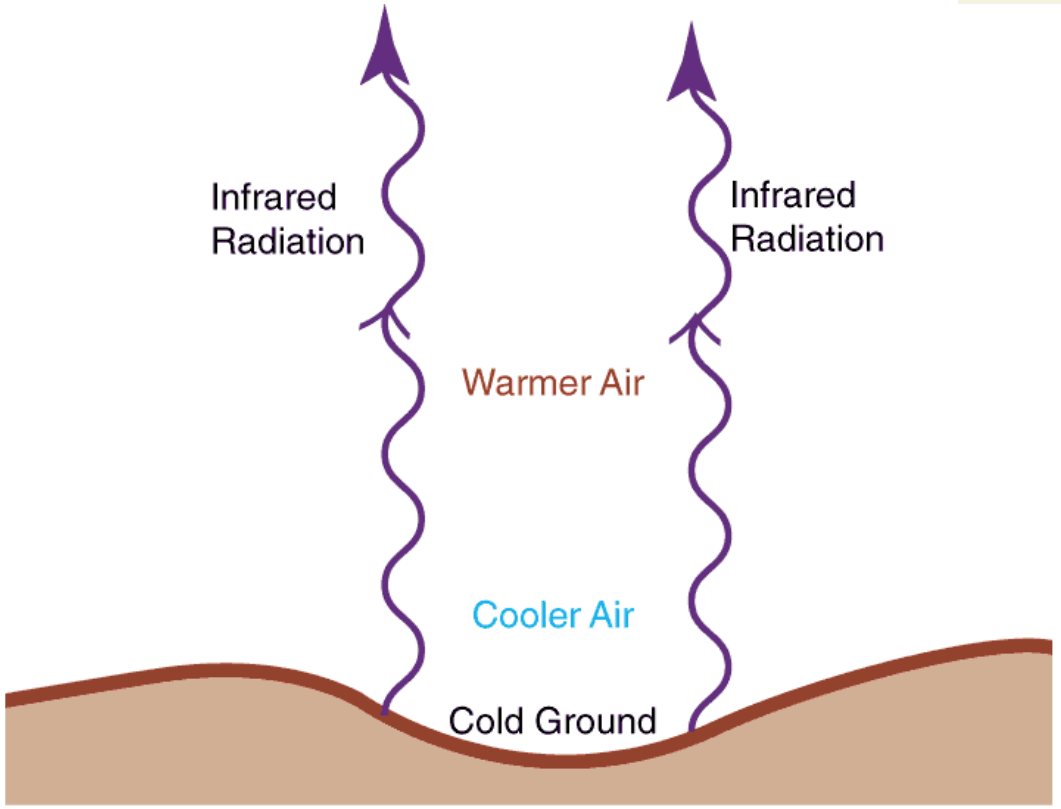


Temperature inversion

Radiation inversion



http://www.smithandgeorg.com.au/articles/agchemnews/november2012/surfacetempinversion.php



http://ffden-2.phys.uaf.edu/212_spring2007.web.dir/Amber_Smith/Pictures/Radiation%20inversion.gif

Temperature inversion

Subsidence inversion

- Caused by the descending air to a high pressure zone

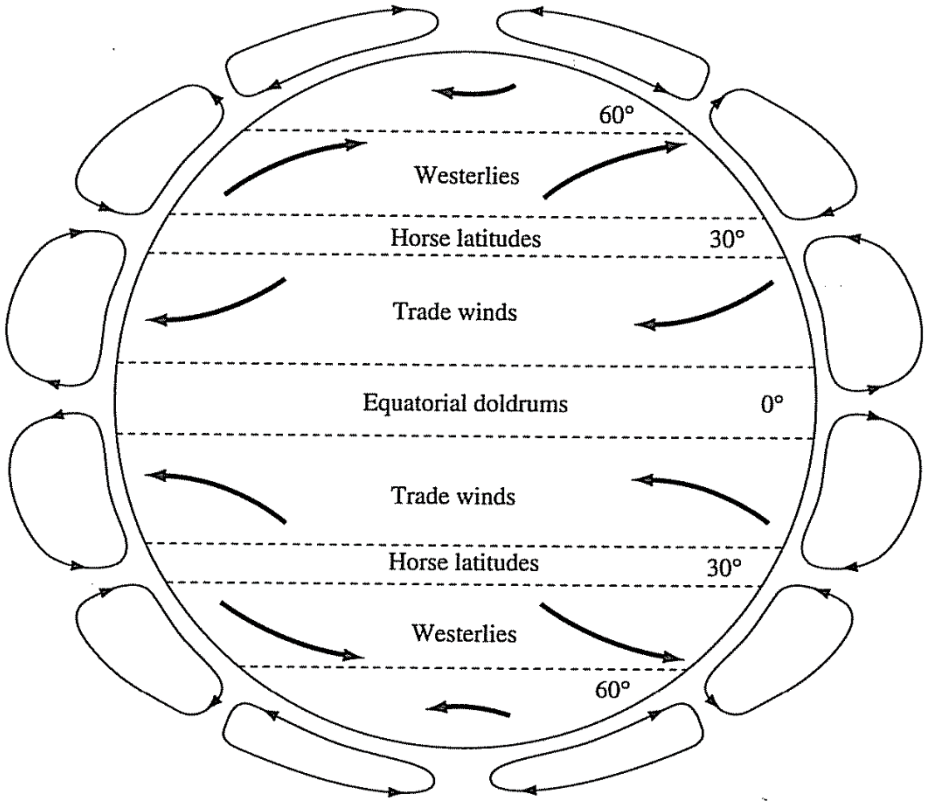


Figure 45

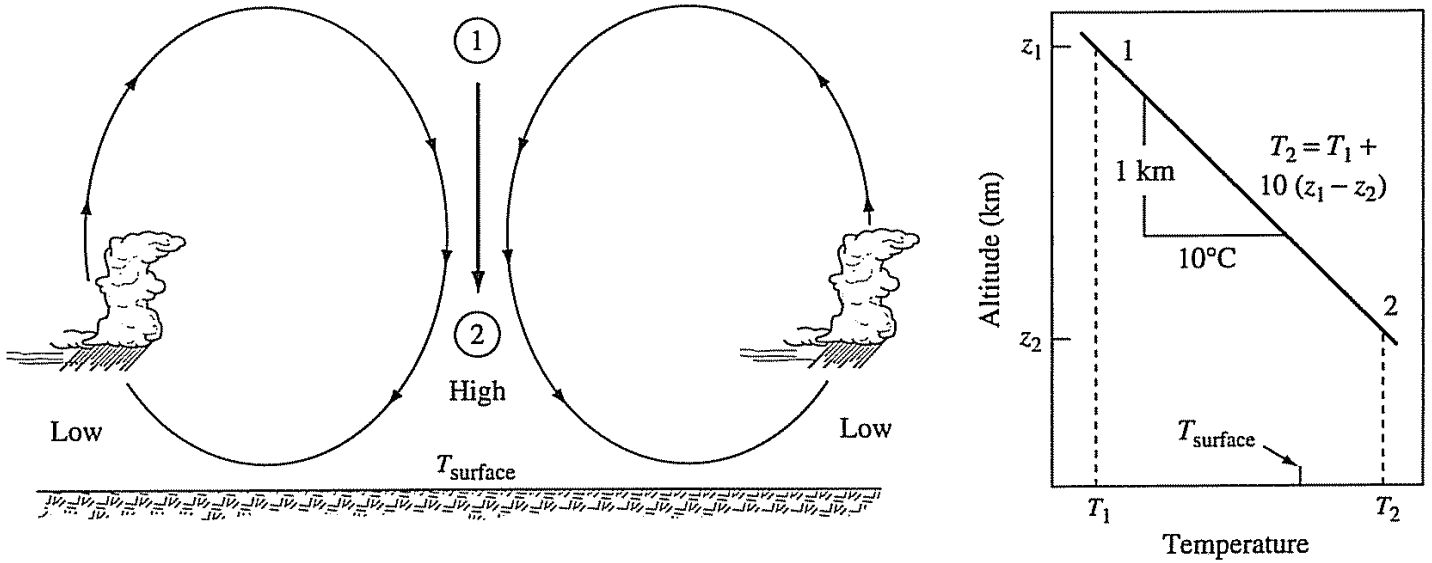


FIGURE 43 Descending air is warmed 10°C/km by compression. It is common to have this subsidence heating create warmer conditions aloft than near the surface, which causes a temperature inversion. Clear skies are typical in a region of high pressure, while cloudiness and precipitation are common in areas with low atmospheric pressure.

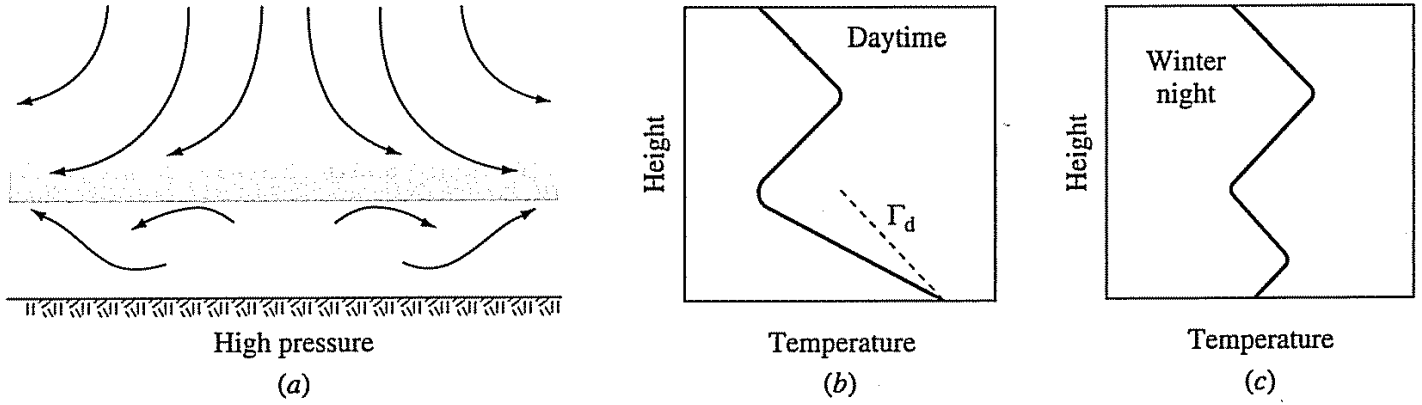


FIGURE 44 Descending air causes subsidence inversions (*a*). During the day, air under the inversion may be unstable due to solar warming of the surface (*b*). Radiation inversions may form under the subsidence inversion when nights are clear, especially in winter (*c*).

Mixing depth

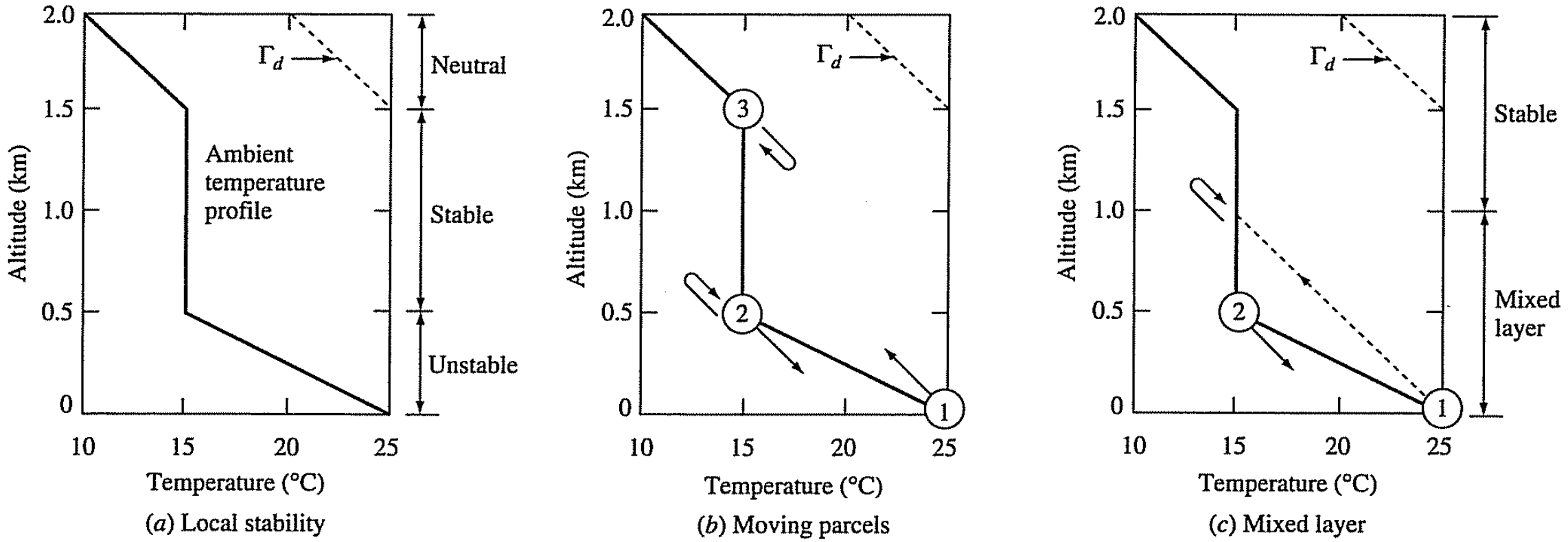
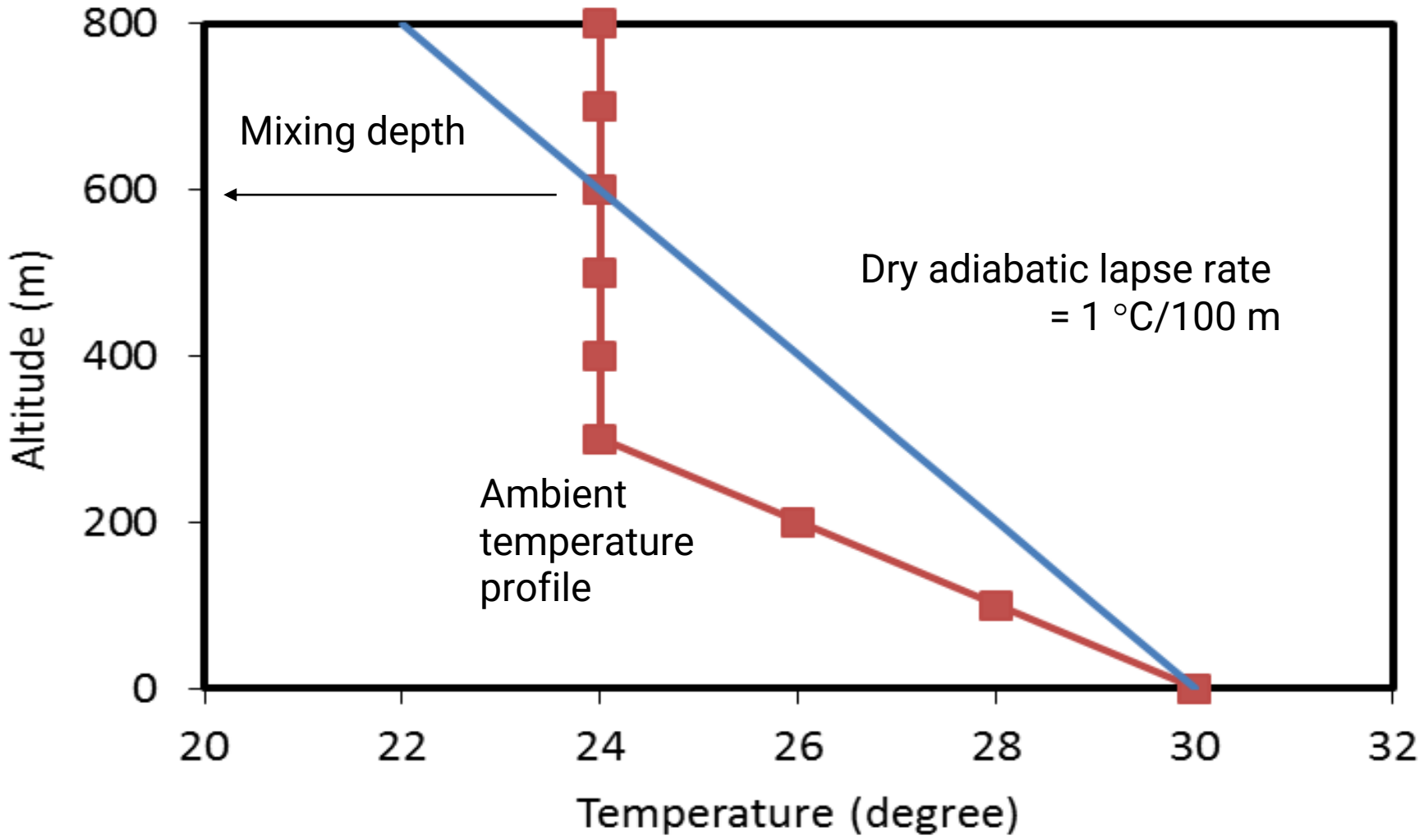
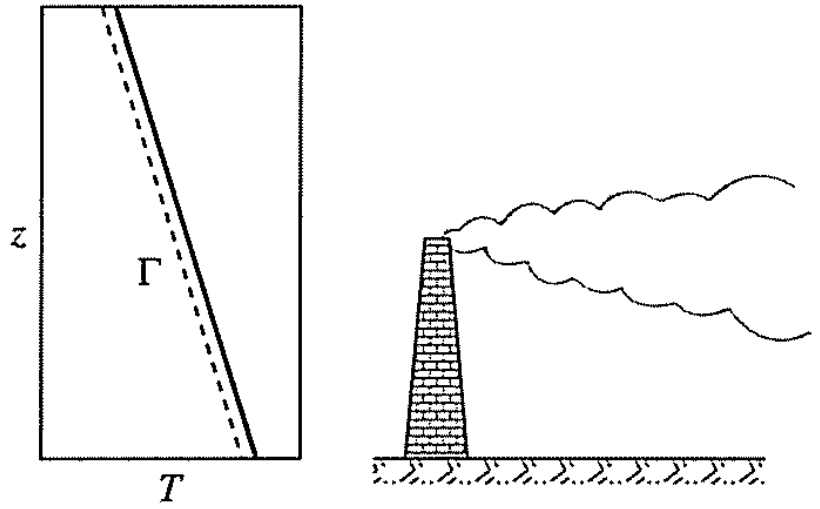


Figure 7.46

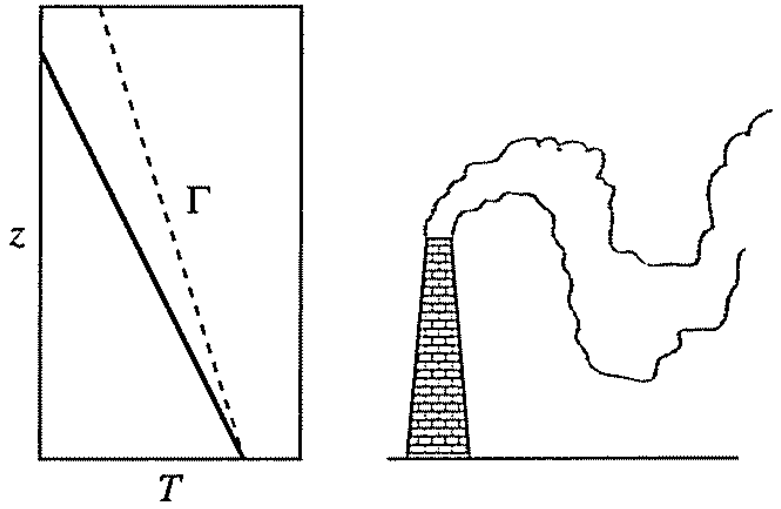
Mixing depth





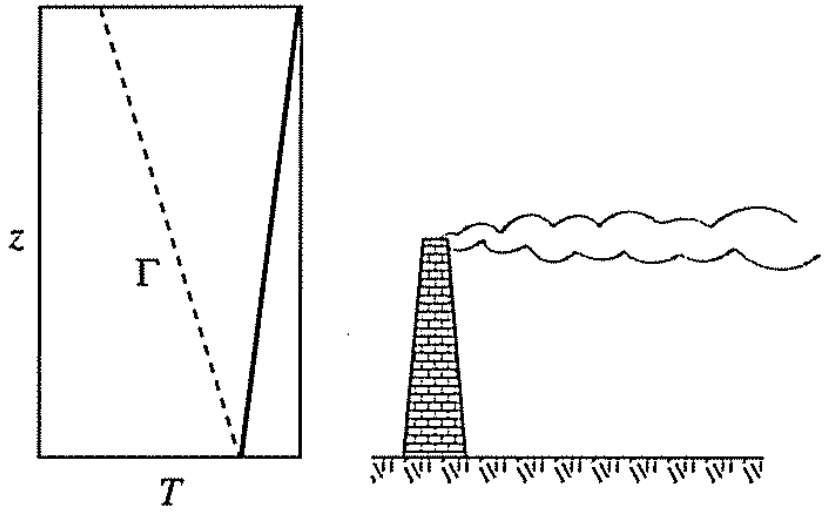
Coning

(*a*)



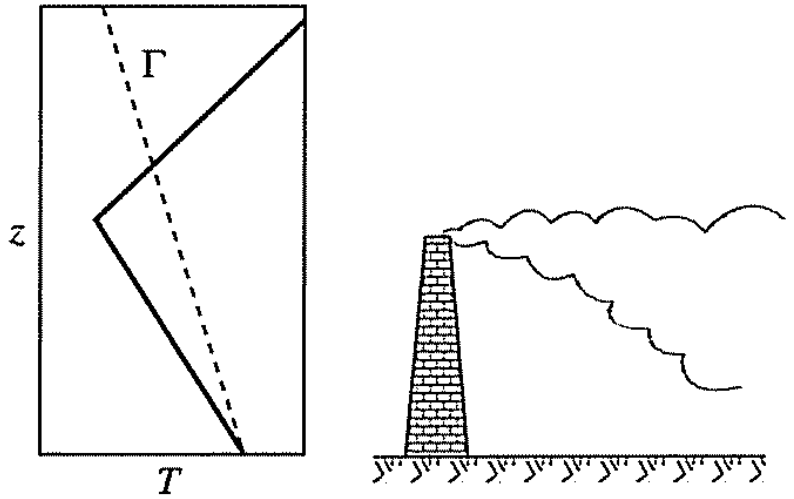
Looping

(*b*)



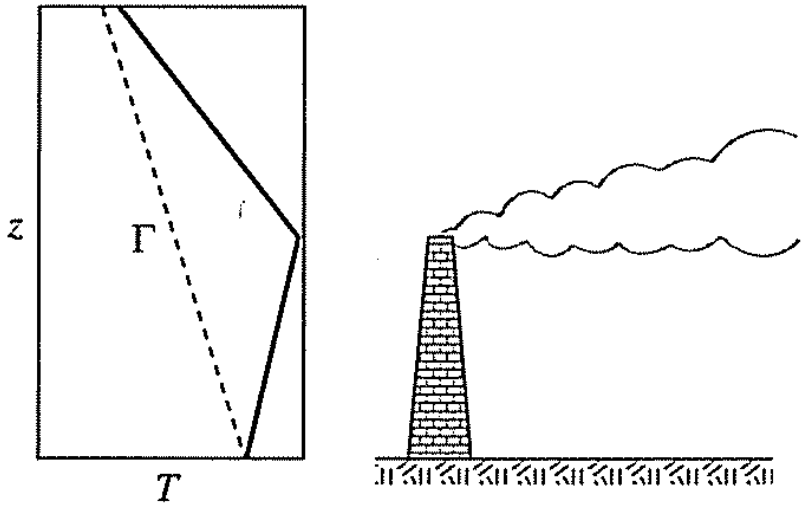
Fanning

(*c*)



Fumigation

(*d*)



Lofting

(*e*)

Point source Gaussian plume model

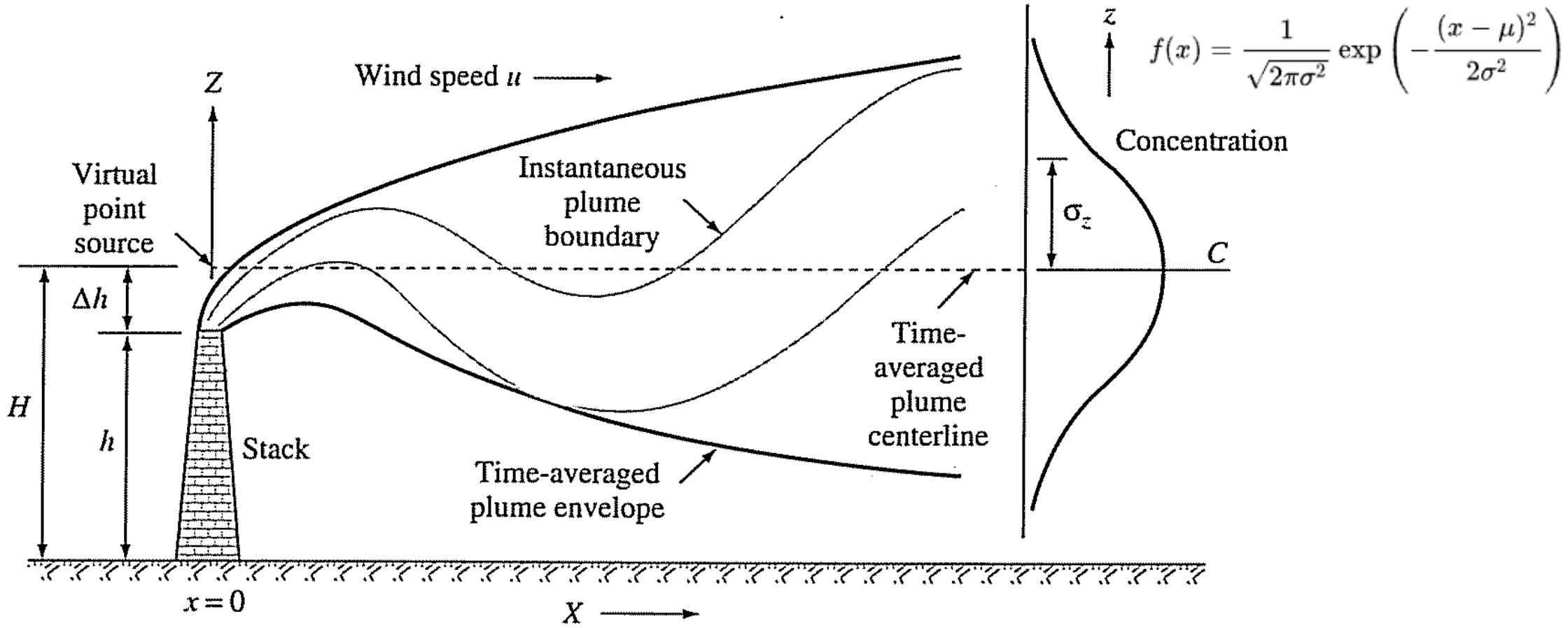
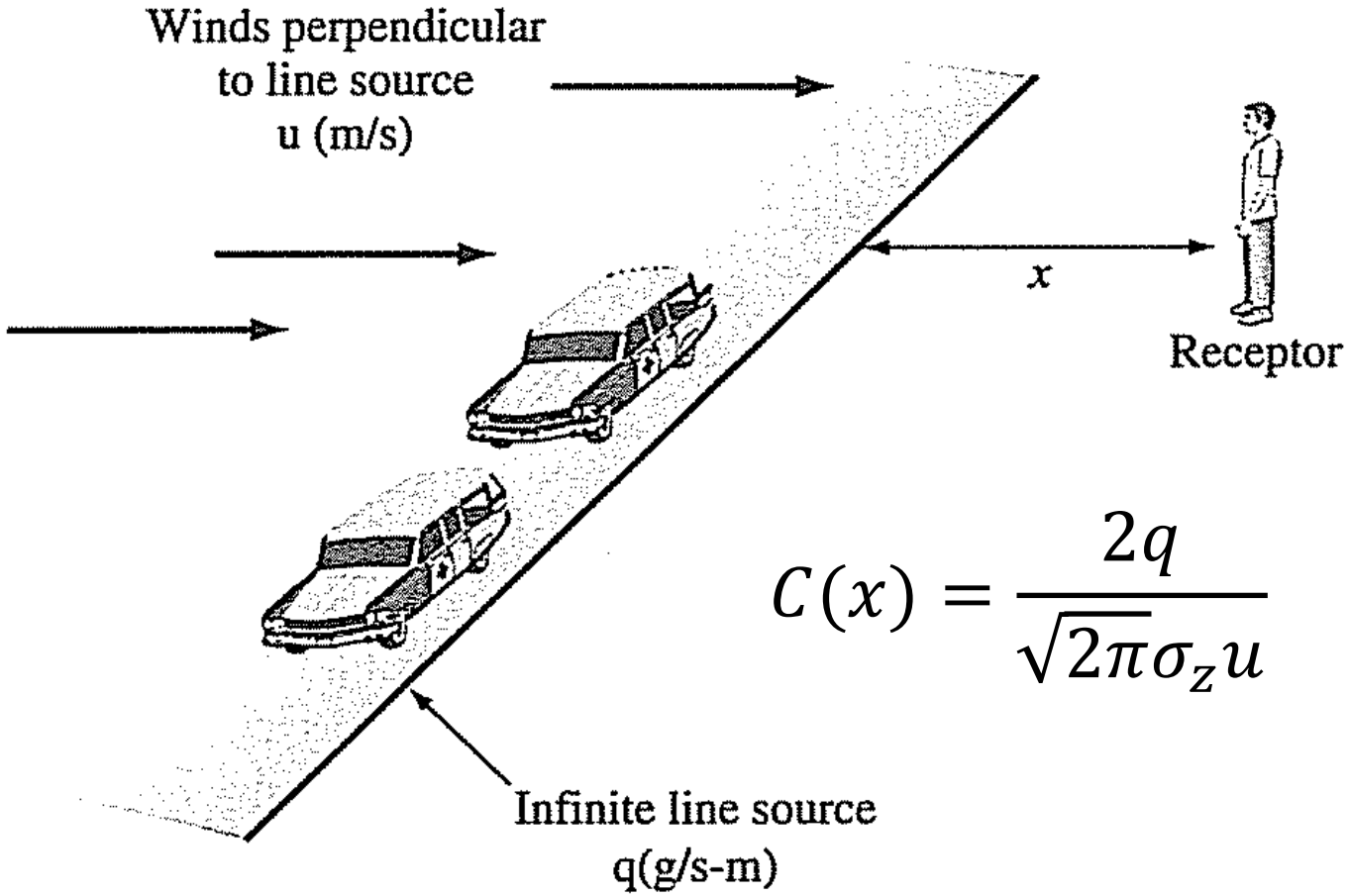


FIGURE 48 The instantaneous plume boundary and a time-averaged plume envelope.

Concentration @ ground level

$$C(x, y) = \frac{Q}{\pi u_H \sigma_y \sigma_z} \exp\left(\frac{-H^2}{2\sigma_z^2}\right) \exp\left(\frac{-y^2}{2\sigma_y^2}\right)$$

Line source model



$$C(x) = \frac{2q}{\sqrt{2\pi}\sigma_z u}$$

Ex. 7.16

10 vehicles/s, each emiting 3.4 g/mile CO; Wind speed = 2.2 m/s

? CO concentration at 100 m away from the road

q = 10/s x 3.4 g/1609m = 0.021 g/s-m

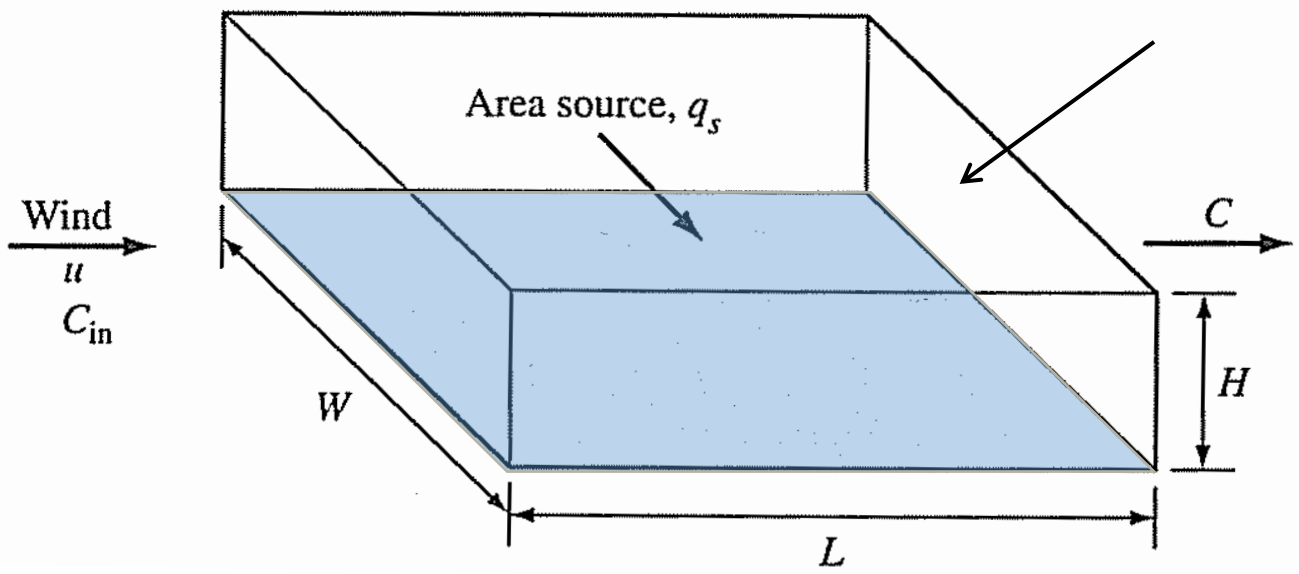
σ_z depends on atmospheric stability condition (assume σ_z = 4.6 m)

→ C(100m) = 1.7 mg/m^3

q: source intensity (g/s/m)
u: wind speed (m/s)
x: distance to the source (m)
σ_z: vertical dispersion parameter (m)

Area source model (box model)

Well mixed, uniform concentration of C



q_s: area source emission rate (g/m^2/s)
H: mixing height (m)
u: wind speed (m/s)
C_in: inlet concentration (g/m^3)
C: pollutant concentration in the box (g/m^3)
L: box length in the wind direction (m)
W: box width (m)

$$q_s WL + C_{in} uWH = CuWH$$

$$C(\text{stead-state}) = \frac{q_s L}{uH} + C_{in} \qquad C(t) = (\frac{q_s L}{uH} + C_{in})(1 - e^{-ut/L})$$

C(t): concentration as a function of time

Example 7.17 Evening Rush Hour Traffic

Suppose within a square city, 15 km on a side, there are 200000 cars on the road, each being driven 30 km between 4 pm and 6 pm, and each emitting 3 g/km of CO. It is a clear winter evening with a radiation inversion that restricts mixing to 20 m. The wind is bringing clean air at a steady rate of 1.0 m/s along an edge of the city. Use a box model to **estimate the CO concentration at 6 pm** if there was no CO in the air at 4 pm, and only source of CO is cars. Assume that CO is conservative and there is complete and instantaneous mixing in the box.

Solution:

The emission per m^2, q_s, would be

$$q_s = \frac{200000 \times 30 \times 3}{15000^2 \times 2 \times 3600} = 1.1 \times 10^{-5} \text{g/s/m}^2$$

The concentration after 2 hours (7200 s) would be

$$C(t) = \frac{q_s L}{uH}\left(1 - e^{-\frac{ut}{L}}\right) = \frac{1.1 \times 10^{-5} \times 15000}{1.0 \times 20}\left(1 - e^{-1.0 \times \frac{7200}{15000}}\right) = 3.2 \text{ mg/m}^3$$

Indoor air quality

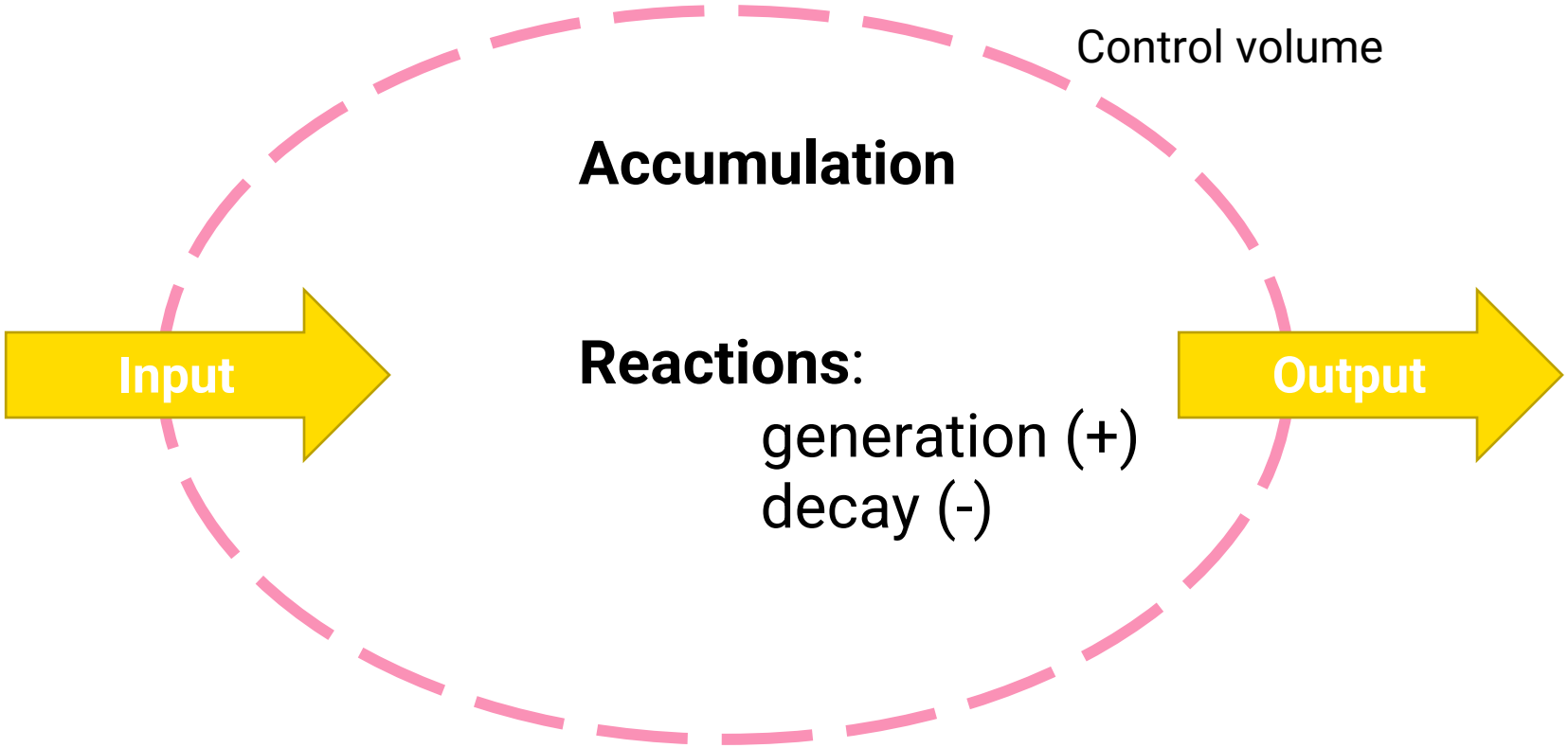
We spent about 90% of our time indoor!

Less dilution compared to outdoor pollutants

Indoor air pollutants

- Cooking → CO, NOx, HC, particulates
- Cigarette smoking → CO, benzene, aldehydes, particulates
- Building materials → VOCs (particularly formaldehyde HCOH)
- Asbestos (石棉)
- Radon (radioactive)

Indoor air quality model (box model)



Accumulation rate = **Input rate** - **Output rate** + **Reaction rate**

Indoor air quality model (box model)

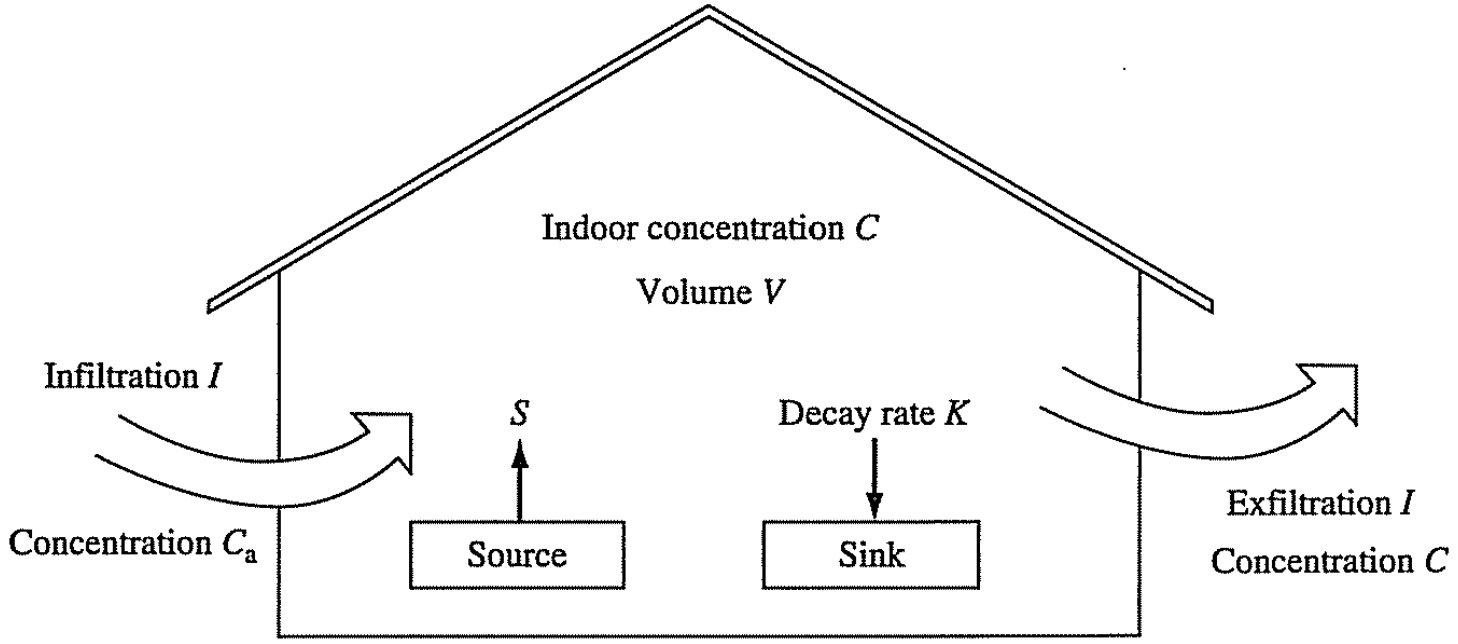


FIGURE 65 Box model for indoor air pollution.

V: indoor volume (m3)
C: indoor pollutant concentration (g/m^3)
Ca: outdoor / ambient pollutant concentration (g/m3))
S: indoor source emission rate (g/s)
I: infiltration / exfiltration flow rate (m3/s)
n: air exchange rate (1/s)
K: first-order decay rate constant (1/s)
t: time (s)

Ventilation is expressed as I=Q=nV

Rate of increase in the box = **Rate of pollution entering the box** - **Rate of pollution leaving the box** - **Rate of decay in the box**

$$V\frac{dC}{dt} = (S + C_a nV) - CnV - KCV$$

Q

Indoor air quality model (box model)

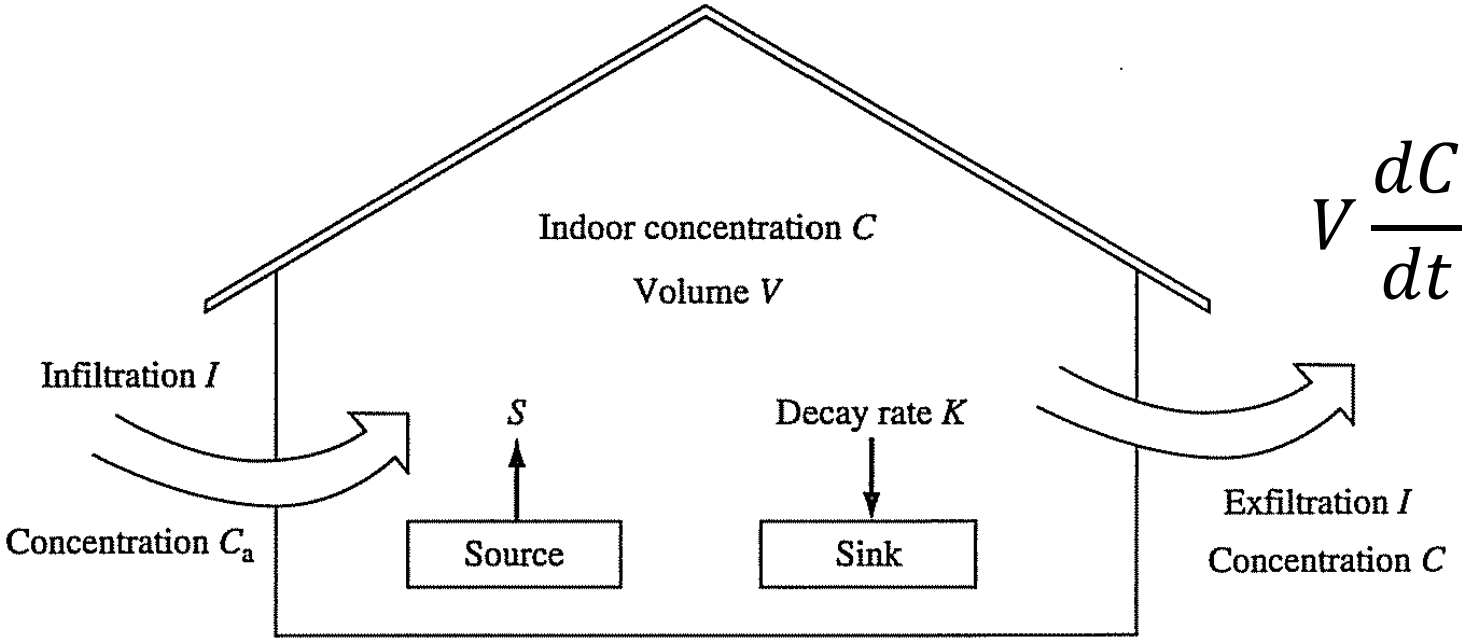


FIGURE 65 Box model for indoor air pollution.

$$V\frac{dC}{dt} = (S + C_a nV) - CnV - KCV$$

With negligible outdoor concentration (Ca = 0), the steady-state equation become:

$$C = \frac{S/V}{n + K}$$

Example 7.19 Indoor Radon Concentration

Suppose the soil under a single-story house emits 1.0 pCi/m^2-s of radon gas. As a worst case, assume that all of this gas finds its way through the floor and into the house. The house has 250 m^2 of floor space, an average ceiling height of 2.6 m, and an air change rate of 0.9 ach. Estimate the steady-state concentration of radon in the house, assuming the ambient concentration is negligible. The decay rate of radon is given in Table 14 as K = 7.6×10^{-3}/hr

Solution:

We can find the steady-state concentration using the box model equation with C_a = 0. which we could have calculated using (3.8) along with radon's 3.8-day half-life:

$$C(\infty) = \frac{S/V}{n+K} = \frac{\frac{1\times 3600\times 250}{250\times 2.6}}{0.9 + 7.6\times 10^{-3}} = 1.5\times 10^{3}\ \text{pCi/m}^3$$

Reading

Reading:

- pp367-370, 376-399, 438-444, 448-451, 467-470, 483-486

Practice problems:

- 7.1, 7.29, 7.49, 7.56